



Final Step - A | Mathematics

Sets, Relations, Statistics & Mathematical Reasoning

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED ‘*’ MAY HAVE MORE THAN ONE CORRECT OPTION.

- If the mean deviation of the numbers $1, 1+d, \dots, 1+100d$ from their mean is 255, then a value of d is :
 (A) 10.1 (B) 20.2 (C) 10 (D) 5.05
- Consider the following two statements :
P : If 7 is an odd number, then is divisible by 2 **Q :** If 7 is a prime number, then is an odd number.
 If V_1 is the value of the contra-positive of P and V_2 is the truth value of contra-positive of Q , then the ordered pair (V_1, V_2) equals :
 (A) (T, T) (B) (T, F) (C) (F, T) (D) (F, F)
- The mean of 5 observations is 5 and their variance is 124. If three of the observations are 1, 2 and 6; then the mean deviation from the mean of the data is :
 (A) 2.4 (B) 2.8 (C) 2.5 (D) 8.38
- The contra-positive of the following statement, “If the side of a square doubles, then its area increases four times”, is :
 (A) If the side of a square is not doubled, then its area does not increase four times
 (B) If the area of a square increases four times, then its side is doubled
 (C) If the area of a square increases four times, then its side is not doubled
 (D) If the area of a square does not increase four times, then its side is not doubled
- In a certain town, 25% of the families own a phone and 15% own a car; 65% families own neither a phone nor a car and 2,000 families own both a car and a phone. Consider the following three statements :
 (a) 5% families own both a car and a phone
 (b) 35% families own either a car or a phone
 (c) 40,000 families live in the town.
 Then,
 (A) Only (a) and (b) are correct (B) Only (a) and (c) are correct
 (C) Only (b) and (c) correct (D) All (a), (b) and (c) are correct
- A factory is operating in two shifts, day and night, 70 and 30 respectively. If per day mean wage of the day shift workers is Rs. 54 and per day mean wage of all the workers in Rs. 60, then per day mean wage of the night shift workers (in Rs.) is :
 (A) 66 (B) 69 (C) 74 (D) 75

7. The contrapositive of the statement "If it is raining, then I will not come" is :
 (A) If I will come, then it is not raining (B) If I will not come, then it is raining
 (C) If I will not come, then it is not raining (D) If I will come, then it is raining
8. Consider the following statements :
 P : Suman is brilliant Q : Suman is rich R : Suman is honest
 The negation of the statement,
 "Suman is brilliant and dishonest if and only if Suman is rich" can be equivalently expressed as :
 (A) $\sim Q \leftrightarrow \sim P \wedge R$ (B) $\sim Q \leftrightarrow \sim P \vee R$ (C) $\sim Q \leftrightarrow P \vee \sim R$ (D) $\sim Q \leftrightarrow P \wedge \sim R$
9. Let P be the relation defined on the set of all real numbers such that :
 $P = \{(a, b) : \sec^2 a - \tan^2 b = 1\}$. Then P is :
 (A) reflexive and symmetric but not transitive (B) reflexive and transitive but not symmetric
 (C) symmetric and transitive but not reflexive (D) an equivalence relation
10. In a set of $2n$ distinct observations, each of the observation below the median of the all observations is increased by 5 and each of the remaining observations is decreased by 3. Then the mean of the new set of observations :
 (A) increases by 1 (B) decreases by 1 (C) decreases by 2 (D) increases by 2
11. The contrapositive of the statement "I go to school if it does not rain" is :
 (A) It is rains, I do not go to school (B) If I do not go to school, it rains
 (C) If it rains, I go to school (D) If I go to school, it rains
12. The proposition $\sim(p \vee \sim q) \vee \sim(p \vee q)$ is logically equivalent to :
 (A) p (B) q (C) $\sim p$ (D) $\sim q$
13. Let $\bar{X}$ and M.D. be the mean and the mean deviation about $\bar{X}$ of n observations $x_i, i = 1, 2, \dots, n$. If each of the observations is increased by 5, then the mean and the deviation about the new mean, respectively, are :
 (A) $\bar{X}, \text{M.D.} + 5$ (B) $\bar{X} + 5, \text{M.D.} + 5$ (C) $\bar{X}, \text{M.D.}$ (D) $\bar{X} + 5, \text{M.D.}$
14. Let p, q, r denote arbitrary statements. Then the logically equivalent of the statement $P \Rightarrow (q \vee r)$ is :
 (A) $(p \Rightarrow q) \wedge (p \Rightarrow \sim r)$ (B) $(p \Rightarrow q) \vee (p \Rightarrow r)$
 (C) $(p \Rightarrow \sim q) \wedge (p \Rightarrow r)$ (D) $(p \vee q) \Rightarrow r$
15. A relation on the set $A = \{x : |x| < 3, x \in Z\}$, where Z is the set of integers is defined by
 $R = \{(x, y) : y = |x|, x \neq -1\}$. Then the number of elements in the power set of R is :
 (A) 64 (B) 8 (C) 16 (D) 32
16. Let $\bar{x}, M$ and σ^2 be respectively the mean, mode and variance of n observations $x_1, x_2, \dots, x_n$ and $d_i = -x_i - a, i = 1, 2, \dots, n$, where a is any number.
Statement I : Variance of $d_1, d_2, \dots, d_n$ is σ^2 .
Statement II : Mean and mode of $d_1, d_2, \dots, d_n$ are $-\bar{x} - a$ and $-M - a$, respectively.
 (A) Statements I and statement II are both false (B) Statements I and statement II are both true
 (C) Statements I is true and statement II is false (D) Statements I is false and statement II is true

17. The contrapositive of the statement “if I am not feeling well, then I will go to the doctor” is :
 (A) If I am feeling well, then I will not go to the doctor
 (B) If I will go to the doctor, then I am feeling well
 (C) If I will not go to the doctor, then I am feeling well
 (D) If I will go to the doctor, then I am not feeling well
18. The mean of a data set consisting of 20 observations is 40. If one observation 53 was wrongly recorded as 33, then the correct mean will be :
 (A) 41 (B) 49 (C) 40.5 (D) 42.5
19. Let $A = \{1, 2, 3, 4\}$ and $R: A \rightarrow A$ be the relation defined by : $R = \{(1, 1), (2, 3), (3, 4), (4, 2)\}$. The correct statement is :
 (A) R does not have an inverse (B) R is not a one to one function
 (C) R is an onto function (D) R is not a function
20. Mean of 5 observations is 7. If four of these observations are 6, 7, 8, 10 and one is missing then the variance of all the five observations is :
 (A) 4 (B) 6 (C) 8 (D) 2
21. Let $R = \{(3, 3), (5, 5), (9, 9), (12, 12), (5, 12), (3, 9), (3, 12), (3, 5)\}$ be a relation on the set $A = \{3, 5, 9, 12\}$. Then, R is :
 (A) reflexive, symmetric but not transitive (B) symmetric, transitive but not reflexive
 (C) an equivalence relation (D) reflexive, transitive but not symmetric
22. The statement $p \rightarrow (q \rightarrow p)$ is equivalent to :
 (A) $p \rightarrow q$ (B) $p \rightarrow (p \vee q)$ (C) $p \rightarrow (p \rightarrow q)$ (D) $p \rightarrow (p \wedge q)$
23. For integers m and n , both greater than 1, consider the following three statements :
 $P: m$ divides n
 $R: m$ is prime
 $Q: m$ divides n^2
 Then :
 (A) $Q \wedge R \rightarrow P$ (B) $P \wedge Q \rightarrow R$ (C) $Q \rightarrow R$ (D) $Q \rightarrow P$
24. If the median and the range of four numbers $\{x, y, 2x + y, x - y\}$, where $0 < y < x < 2y$, are 10 and 28 respectively, then the mean of the numbers is :
 (A) 18 (B) 10 (C) 5 (D) 14
25. Let $R = \{(x, y): x, y \in N \text{ and } x^2 - 4xy + 3y^2 = 0\}$, where N is the set of all natural numbers. Then the relation R is :
 (A) reflexive but neither symmetric nor transitive
 (B) symmetric and transitive
 (C) reflexive and symmetric
 (D) reflexive and transitive

26. The Boolean Expression $(p \wedge \sim q) \vee q \vee (\sim p \wedge q)$ is equivalent to :
 (A) $p \wedge q$ (B) $p \vee q$ (C) $p \vee \sim q$ (D) $\sim p \wedge q$
27. The negation of $\sim s \vee (\sim s \wedge r)$ is equivalent to :
 (A) $s \vee (r \vee \sim s)$ (B) $s \wedge r$ (C) $s \wedge \sim r$ (D) $s \wedge (r \wedge \sim s)$
28. The mean of the data set comprising of 16 observations is 16. If one of the observation valued 16 is deleted and three new observation valued 3, 4 and 5 are added to the data, then the mean of the resultant data is :
 (A) 15.8 (B) 14.0 (C) 16.8 (D) 16.0
29. If $X = \left\{ \left(4^n - 3n - 1 : n \in N \right) \right\}$ and $Y = \{ 9(n-1) : n \in N \}$, when N is the set of natural numbers, then $X \cup Y$ is equal to :
 (A) $Y - X$ (B) X (C) Y (D) N
30. The variance of first 50 even natural numbers is :
 (A) 833 (B) 437 (C) $\frac{437}{4}$ (D) $\frac{833}{4}$
31. The statement $\sim (p \leftrightarrow \sim q)$ is :
 (A) equivalent to $\sim p \leftrightarrow q$ (B) a tautology
 (C) a fallacy (D) equivalent to $p \leftrightarrow q$
32. Consider :
Statement I : $(p \wedge \sim q) \wedge (\sim p \wedge q)$ is a fallacy.
Statement II : $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ is a tautology.
 (A) Statement I is false; Statement II is true
 (B) Statement I is true; Statement II is true; Statement II is a correct explanation for Statement I
 (C) Statement I is true; Statement II is true; Statement II is not a correct explanation for Statement I
 (D) Statement I is true; Statement II is false
33. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is :
 (A) 211 (B) 256 (C) 220 (D) 219

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

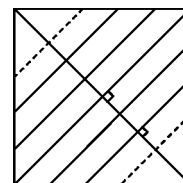
34. If $aN = \{ax : x \in N\}$, then the set $3N \cap 7N$ will be multiple of:
35. In a statistical investigation of 1003 families of Calcutta, it was found that 63 families has neither a radio nor a T.V., 794 families has a radio and 187 has T.V. The number of families in that group having both a radio and a T.V. is:
36. If two sets A and B are having 99 elements in common, then the number of elements common to each of the sets $A \times B$ and $B \times A$ is $(k)^2$ then k will be:
37. A survey shows that 63% of of the Americans like cheese whereas 76% like apples. If $x\%$ of the Americans like both cheese and apples, then a value of x can be:
38. If $n(A) = 3$, $n(B) = 6$ and $A \subseteq B$. Then the number of elements in $A \cup B$ is equal to:
39. The number of proper subsets of the set $\{1, 2, 3\}$ is:
40. If $n(A) = 4$, $n(B) = 3$, $n(A \times B \times C) = 24$, then $n(C) =$
41. The number of elements in the set $\{(a, b) : 2a^2 + 3b^2 = 35a, a, b \in Z\}$, where Z is the set of all integers, is:
42. Given $n(U) = 20$, $n(A) = 12$, $n(B) = 9$, $n(A \cap B) = 4$, where U is the universal set, A and B are subsets of U, then $n((A \cup B)^C) =$
43. In a class of 30 pupils, 12 take needle work, 16 take physics and 18 take history. If all the 30 students take at least one subject and no one takes all three then the number of pupil taking 2 subjects is:
44. A student obtains 75%, 80% and 85% in three subjects. If the marks of another subject is added, then his average cannot be less than $K\%$ then K is:
45. The mean of 100 items is 49. It was discovered that three items which should have been 60, 70, 80 were wrongly read as 40, 20, 50, respectively. The correct mean is:
46. The mean weight per student in a group of seven students is 55 kg. If the individual weights of six students are 52, 58, 55, 53, 56 and 54, then the weight of the seventh student is m kg, then m is:
47. Let $x_1, x_2, \dots, x_n$ be n observations such that $\sum x_i^2 = 400$ and $\sum x_i = 80$. Then least possible value of ' n ' is:
48. If in a moderately asymmetrical distribution the mode and the mean of the data are 6λ and 9λ , respectively, then the median is $K\lambda$ then K is:

Sequence & Series

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT.

- If the p th and q th terms of a GP are q and p , respectively, then $(p + q)$ th term is :
 (A) $\left(\frac{q^p}{p^q}\right)^{\frac{1}{p-q}}$ (B) $\left(\frac{q^q}{p^p}\right)^{\frac{1}{p-q}}$ (C) $\left(\frac{p^q}{q^p}\right)^{\frac{1}{p-q}}$ (D) None of these
- The sum of the geometric progression 0.15, 0.015, 0.0015, ... 20 terms is :
 (A) $\frac{1}{6}[1 - (0.1)^{20}]$ (B) $[1 - (0.1)^{20}]$ (C) $\frac{1}{6}[1 + (0.1)^{20}]$ (D) None of these
- If 100 times the 100th term of an AP with non-zero common difference equals the 50 times its 50th term, then the 150th term of this AP is :
 (A) -150 (B) 150 times its 50th term
 (C) 150 (D) Zero
- If a_1, a_2, a_3, a_4 are in HP, then $\frac{1}{a_1 a_4} \sum_{r=1}^3 a_r a_{r+1}$ is a root of :
 (A) $x^2 + 2x + 15 = 0$ (B) $x^2 + 2x - 15 = 0$ (C) $x^2 - 2x + 15 = 0$ (D) $x^2 - 2x - 15 = 0$
- If $a_1, a_2, a_3, \dots, a_n$ are in HP and $f(k) = \sum_{r=1}^n a_r - a_k$, then $\frac{a_1}{f(1)}, \frac{a_2}{f(2)}, \frac{a_3}{f(3)}, \dots, \frac{a_n}{f(n)}$ are in :
 (A) AP (B) GP (C) HP (D) None of these
- Let a_n be the n th term of the GP of positive numbers. Let $\sum_{n=1}^{100} a_{2n} = \alpha$ and $\sum_{n=1}^{100} a_{2n-1} = \beta$, such that $\alpha \neq \beta$, then the common ratio is :
 (A) α / β (B) β / α (C) $\sqrt{(\alpha / \beta)}$ (D) $\sqrt{(\beta / \alpha)}$
- The sum of the product taken two at a time of the numbers 1, 2, $2^2, 2^3, \dots, 2^{n-2}, 2^{n-1}$ is :
 (A) $\frac{1}{3} \cdot 2^{2n} + \frac{2}{3}$ (B) $\frac{1}{3} \cdot 2^{2n} - 2^n + \frac{1}{3}$ (C) $\frac{1}{3} \cdot 2^{2n} - \frac{1}{3}$ (D) $\frac{1}{3} \cdot 2^{2n} - 2^n + \frac{2}{3}$
- In an AP, $a_7 = 9$ and $a_1 a_2 a_7$ is least, then the common difference is :
 (A) $\frac{13}{20}$ (B) $\frac{23}{20}$ (C) $\frac{33}{20}$ (D) $\frac{43}{20}$
- Value of $L = \lim_{n \rightarrow \infty} \frac{1}{n^4} \left[1 \cdot \left(\sum_{k=1}^n k \right) + 2 \left(\sum_{k=1}^{n-1} k \right) + 3 \left(\sum_{k=1}^{n-2} k \right) + \dots + n \cdot 1 \right]$ is :
 (A) 1/24 (B) 1/12 (C) 1/6 (D) 1/3
- Three numbers a, b, c between 2 and 18 such that
 (i) $a + b + c = 25$ (ii) 2, a, b are consecutive terms of an AP
 (iii) $b, c, 18$ are consecutive terms of a GP
 If $G = \text{Max}\{a, b, c\}$ and $L = \text{Min}\{a, b, c\}$ then the value of $G + L$ is :
 (A) 17 (B) 18 (C) 20 (D) 34

11. If t_r denotes the r^{th} term of an G.P and $t_1 t_5 t_{10} t_{16} t_{21} t_{25} = 512$, then the value of its 13^{th} term is :
 (A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) 2 (D) $2\sqrt{2}$
12. If $(1+3+5+7+\dots+(2p-1)) + (1+3+5+\dots+(2q-1)) = 1+3+5+\dots+(2r-1)$, then least possible value of $p+q+r$ (Given $p > 5$) is :
 (A) 24 (B) 32 (C) 27 (D) 25
13. If $f(n) = 1 \times 2 \times 3 \times \dots \times (n-1) \times n$, where n is a natural number, then $\sum_{r=1}^{20} r f(r) = f(x) - 1$, then the value of x is :
 (A) 25 (B) 21 (C) 28 (D) 24
14. Coefficient of x^0 in $(x-a)(x-b)(x-c)(x-d)(x-e)\dots(x-x_0)$ is :
 (A) Product of roots (B) Sum of roots (C) 1
 (D) Depends on roots and number of roots
15. If $x = \sum_{n=0}^{\infty} a^n$, $y = \sum_{n=0}^{\infty} b^n$, $0 < a < b < 1$ and $z = \sum_{n=0}^{\infty} \left(\frac{a}{b}\right)^n$, then :
 (A) $x + yz = x(y+z)$ (B) $xyz = x + y + z$ (C) $xy + z = y(x+z)$ (D) None of these
16. If $(1.05)^{50} = 11.658$, then $\sum_{n=1}^{49} (1.05)^n$ equals :
 (A) 208.34 (B) 212.12 (C) 212.16 (D) 213.16
17. Number of common terms of two sequences 17, 21, 25, ... 417 and 16, 21, 26, ... 466 are :
 (A) 20 (B) 19 (C) 21 (D) 22
18. Let $t_r = 2^{r/2} + 2^{-r/2}$. Then $\sum_{r=1}^{10} t_r^2$ is equal to :
 (A) $\frac{2^{21}-1}{2^{10}} + 20$ (B) $\frac{2^{21}-1}{2^{10}} + 19$ (C) $\frac{2^{21}-1}{2^{20}} - 1$ (D) None of these
19. Value of $\frac{1}{\sqrt{3}+\sqrt{7}} + \frac{1}{\sqrt{7}+\sqrt{11}} + \frac{1}{\sqrt{11}+\sqrt{15}} + \dots n$ terms, is equal to :
 (A) $\frac{1}{4}(\sqrt{4n+3} - \sqrt{3})$ (B) $\frac{1}{4}(\sqrt{4n+3} + \sqrt{3})$ (C) $\frac{1}{2}(\sqrt{4n+3} - \sqrt{3})$ (D) $\frac{1}{2}(\sqrt{4n+3} + \sqrt{3})$
20. If a, b, c are three distinct positive real numbers, then :
 (A) $(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) > 9$ (B) $(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) < 9$
 (C) $(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$ (D) $(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \leq 9$
21. In the given square, a diagonal is drawn, and parallel line segments joining points on the adjacent sides are drawn on both sides of the diagonal. The length of the diagonal is $n\sqrt{2}$ cm. If the distance between consecutive line segments be $\frac{1}{\sqrt{2}}$ cm, then the sum of the lengths of all possible line segments and the diagonal is:
 (A) $n(n+1)\sqrt{2}$ cm (B) n^2 cm (C) $n(n+2)$ cm (D) $n^2\sqrt{2}$ cm



22. If $1^2 + 2^2 + 3^2 + \dots + 2003^2 = (2003)(4007)(334)$ and $(1)(2003) + (2)(2002) + (3)(2001) + \dots + (2003)(1) = (2003)(334)(x)$, then x equals :
- (A) 2005 (B) 2004 (C) 2003 (D) 2001
23. In a triangle ABC , which of the following holds goods :
- (A) $(a+b+c)^3 \geq 27(b+c-a)(c+a-b)(a+b-c)$
 (B) $(a+b+c)^3 > 27(b+c-a)(c+a-b)(a+b-c)$
 (C) $(a+b+c)^3 > \frac{27}{8}(b+c-a)(c+a-b)(a+b-c)$
 (D) None of these
24. Consider the A.P. $a_1, a_2, \dots, a_n, \dots$ the G.P., $b_1, b_2, \dots, b_n, \dots$ such that $a_1 = b_1 = 1$; $a_9 = b_9$ and $\sum_{r=1}^9 a_r = 369$, then :
- (A) $b_6 = 27$ (B) $b_7 = 27$ (C) $b_8 = 81$ (D) $b_9 = 18$
25. Let x, y, z be positive real numbers such that $x+y+z = 12$ and $x^3 y^4 z^5 = (0.1)(600)^3$. Then $x^3 + y^3 + z^3$ is equal to :
- (A) 270 (B) 258 (C) 342 (D) 216
26. Let $a_1, a_2, a_3, \dots, a_n, \dots$ be in A.P. If $a_3 + a_7 + a_{11} + a_{15} = 72$, then the sum of first 17 terms is equal to :
- (A) 306 (B) 153 (C) 612 (D) 204
27. The sum $\sum_{r=1}^{10} (r^2 + 1) \times (r!)$ is equal to :
- (A) $(11)!$ (B) $10 \times (11!)$ (C) $101 \times (10!)$ (D) $11 \times (11!)$
28. The value of $\sum_{r=16}^{30} (r+2)(r-3)$ is equal to :
- (A) 7785 (B) 7780 (C) 7775 (D) 7770
29. Given an A.P. whose terms are all positive integers. The sum of its first nine terms is greater than 200 and less than 220. If the second term in it is 12, then its 4th term is :
- (A) 8 (B) 16 (C) 20 (D) 24
30. If the sum $\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \dots$ up to 20 terms is equal to $\frac{k}{21}$, then k is equal to :
- (A) 120 (B) 180 (C) 240 (D) 60
31. In a geometric progression, if the ratio of the sum of first 5 terms to the sum of their reciprocals is 49, and the sum of the first and the third term is 35. Then the first term of this geometric progression is :
- (A) 7 (B) 21 (C) 28 (D) 42
32. The sum of the first 20 terms common between has series $3+7+11+15+\dots$ and $1+6+11+16+\dots$ is :
- (A) 4000 (B) 4020 (C) 4200 (D) 4220
33. Let G be the geometric mean of two positive numbers a and b , and M be the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$. If $\frac{1}{M} : G$ is 4 : 5, then $a : b$ can be :
- (A) 1 : 2 (B) 2 : 3 (C) 1 : 4 (D) 3 : 4

34. The least positive integer n such that $1 - \frac{2}{3} - \frac{2}{3^2} - \dots - \frac{2}{3^{n-1}} < \frac{1}{100}$, is :
 (A) 5 (B) 4 (C) 7 (D) 6
35. The number of terms in an A.P. is even; the sum of the odd terms in it is 24 and that the even terms is 30. If the last terms exceeds the first term by $10\frac{1}{2}$, then the number of terms in the A.P. is :
 (A) 4 (B) 8 (C) 12 (D) 16
36. Let $a_1, a_2, a_3, \dots$ be an A.P. such that $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + a_3 + \dots + a_q} = \frac{p^3}{q^3}$; $p \neq q$. Then $\frac{a_6}{a_{21}}$ is equal to :
 (A) 41/11 (B) 121/1681 (C) 11/41 (D) 121/1861
37. The sum of the series $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots$ upto 10 terms is :
 (A) $\frac{18}{11}$ (B) $\frac{22}{13}$ (C) $\frac{20}{11}$ (D) $\frac{16}{9}$
38. Given sum of the first n terms of an A.P. is $2n + 3n^2$. Another A.P. is formed with the same first term and double of the common difference, the sum of n terms of the new A.P. is :
 (A) $n + 4n^2$ (B) $6n^2 - n$ (C) $n^2 + 4n$ (D) $3n + 2n^2$
39. The sum $\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \dots$ up to 11-terms is :
 (A) $\frac{7}{2}$ (B) $\frac{11}{4}$ (C) $\frac{11}{2}$ (D) $\frac{60}{11}$
40. The sum of the series $(2)^2 + 2(4)^2 + 3(6)^2 + \dots$ upto 10 terms is :
 (A) 11300 (B) 11200 (C) 12100 (D) 12300
41. Let the sum of the first three terms of an A.P. be 39 and the sum of its last four terms be 178. If the first term of this A.P. is 10, then the median of the A.P. is :
 (A) 26.5 (B) 28 (C) 29.5 (D) 31
42. If $\sum_{n=1}^5 \frac{1}{n(n+1)(n+2)(n+3)} = \frac{k}{3}$, then k is equal to :
 (A) $\frac{55}{336}$ (B) $\frac{17}{105}$ (C) $\frac{1}{6}$ (D) $\frac{19}{112}$
43. The sum of the 3rd and 4th terms of a G.P. is 60 and the product of its first three terms is 1000. If the first term of this G.P. is positive, then its 7th term is :
 (A) 7290 (B) 320 (C) 640 (D) 2430
44. If $a_1, a_2, a_3, \dots, a_n, \dots$ are in A.P. such that $a_4 - a_7 + a_{10} = m$, then the sum of first 13 terms of this A.P., is :
 (A) 10 m (B) 12 m (C) 13 m (D) 15 m
45. If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots$, is $\frac{16}{5}m$, then m is equal to :
 (A) 101 (B) 100 (C) 99 (D) 102

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

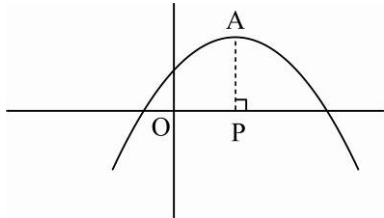
46. The number of terms of an A.P. is even; the sum of the odd terms is 24, sum of the even terms is 30, and the last term exceeds the first by $10\frac{1}{2}$, then the number of terms in the series is:
47. In an increasing G.P., the sum of the first and last term is 66, the product of the second and the last terms is 128 and the sum of the terms is 126. Then number of terms in the series is:
48. If a, b, c, d are in H.P., then value of $\frac{a^{-2} - d^2}{b^{-2} - c^{-2}}$ is:
49. If $S = \frac{2^2 - 1}{2} + \frac{3^2 - 2}{6} + \frac{4^2 - 3}{12} + \dots$ upto 10 terms, then $[S]$ is equal to: (where $[]$ the G.I.F.)
50. If a, b, c are in G.P. and x, y , respectively be arithmetic means between a, b and b, c then the value of $\frac{a}{x} + \frac{c}{y}$ is:
51. Minimum value of $\frac{b+c}{a} + \frac{c+a}{b} + \frac{a+b}{c}$, (for real positive numbers, a, b, c) is:
52. If a, b, c are the sides of a triangle, then the minimum value of $\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c}$ is equal to:
53. The value of $\sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1 = 220$, then the value of n equals:
54. Let a sequence of number be as follows:
- | | | | | | |
|----|----|----|----|----|----|
| | | 1 | | | |
| | | 3 | | 5 | |
| | 7 | | 9 | | 11 |
| | 13 | | 15 | | 17 |
| 21 | | 23 | | 25 | |
| | | | | 27 | |
| | | | | | 29 |
| | | | | | |
| | | | | | |
-
-
- If t_n is the first term of n th row then $\lim_{n \rightarrow \infty} (\sqrt{t_n} - n)$ is equal to: S ; then $2|S|$ will be:
55. Let $U_n = \frac{(n+1)!}{(n+3)!}$ $n \in N$. If $S_n = \sum_{n=1}^n U_n$, then $\lim_{n \rightarrow \infty} 3S_n$ equals:
56. Let a_1, a_2, a_3, a_4 and a_5 be a five term geometric sequence satisfying the condition $0 < a_1 < a_2 < a_3 < a_4 < a_5 < 100$, where each term is an integer. Then the number of such five terms geometric progression is:
57. If a, b, c are in G.P., $a-b, c-a, b-c$ are in H.P., then $a+4b+c$ is equal to:
58. $S_n = \frac{1}{1^3} + \frac{1+2}{1^3+2^3} + \dots + \frac{1+2+\dots+n}{1^3+2^3+\dots+n^3}$; $n=1, 2, 3, \dots$ then S_n is not greater than:
59. If x, y, z and w are positive integers such that $x+2y+3z+4w=50$, then maximum value of $\left(\frac{x^2 y^4 z^3 w}{16}\right)^{1/10}$ is:
60. Let $f(x) = (a^2 + b^2 - 4a - 6b + 13)(2x^2 - 4x + 5)$, where $a, b \in R$ such that $f(0) = f(1) = f(2)$.
- If $a, A_1, A_2, \dots, A_{10}, b$ is an A.P. and $a, H_1, H_2, \dots, H_{10}, b$ is a H.P., then the value of $\frac{1}{10} \left(\sum_{r=4}^8 A_r H_{11-r} \right)$ is:

Quadratic Equations & Inequations

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

- *1. Let $a > 0, b > 0$ and $c > 0$. Then both the roots of the equation $ax^2 + bx + c = 0$ can :
- (A) Be real and negative (B) Have negative real parts
(C) Be rational numbers (D) None of these
2. If $x^2 - (a-3)x + a = 0$ has at least one positive root then :
- (A) $a \in (-\infty, 0) \cup [7, 9]$ (B) $(-\infty, 0) \cup [7, \infty]$
(C) $a \in (-\infty, 0) \cup [9, \infty)$ (D) None of these
3. If $x \in R$, then the expression $9^x - 3^x + 1$ assumes :
- (A) All real values (B) All real values greater than 0
(C) All real values greater than $3/4$ (D) All real values greater than $1/4$
4. For all $x \in R$. The values between which the expression $\frac{9 \cdot 3^{2x} - 6 \cdot 3^x + 4}{9 \cdot 3^{2x} + 6 \cdot 3^x + 4}$ lies are :
- (A) 3^{-1} and 1 (B) -2 and 0 (C) -1 and 1 (D) None of these
- *5. If $a < b < c < d$, then for any real non-zero λ , the quadratic equation $(x-a)(x-c) + \lambda(x-b)(x-d) = 0$, has :
- (A) Non-real roots (B) One real root between a and c
(C) One real root between b and d (D) Irrational roots
- *6. Equation $\frac{\pi^e}{x-e} + \frac{e^\pi}{x-\pi} + \frac{\pi^\pi + e^e}{x-\pi-e} = 0$ has :
- (A) One real root in (e, π) and other in $(\pi-e, e)$
(B) One real root in (e, π) and other in $(\pi, \pi+e)$
(C) Two real roots in $(\pi-e, \pi+e)$
(D) No real root
7. If $0 < a < b < c$, and the roots, α, β of the equation $ax^2 + bx + c = 0$ are non real complex roots, then :
- (A) $|\alpha| = |\beta|$ (B) $|\alpha| = 1$ (C) $|\beta| = 1$ (D) None of these
- *8. If $a, b, c \in R$ and the equality $ax^2 - bx + c = 0$ has complex roots which are reciprocal of each other, then one has :
- (A) $|b| \leq |a|$ (B) $|b| \leq |c|$ (C) $a = c$ (D) $b \leq a$
9. If for all real x , $\frac{x+a}{x^2+bx+c^2}$ cannot lie between two limits, then :
- (A) $b^2 > 4c^2$ and $a^2 + c^2 > ab$ (B) $b^2 > 4c^2$ and $a^2 + c^2 < ab$
(C) $b^2 < 4c^2$ and $a^2 + c^2 > ab$ (D) $b^2 < 4c^2$ and $a^2 + c^2 < ab$

10. If $f(x) = \frac{ax^2 + x - 2}{a + x - 2x^2}$ has the range R when $x \in R$, then :
- (A) $\alpha \in \left(-\frac{1}{8}, 1\right]$ (B) $\alpha \in \left(-\frac{1}{8}, 3\right]$ (C) $a \in [1, 3]$ (D) None of these
11. If $x \in R$, then $f(x) = (x-1)^2 + (x-2)^2 + (x-3)^2 + (x-4)^2$ assumes its minimum value at :
- (A) $x=10$ (B) $x=2.5$ (C) $x=20$ (D) $x=40$
12. The number of real roots of the equation $x^4 + \sqrt{x^4 + 20} = 22$ is :
- (A) 4 (B) 2 (C) 0 (D) 1
13. Let $f(x)$ be a polynomial for which the remainders when divided by $x-1$, $x-2$, $x-3$ respectively are 3, 7, 13. Then the remainder of $f(x)$ when divided by $(x-1)(x-2)(x-3)$ is :
- (A) $f(x)$ (B) $x^2 + x + 1$ (C) $x^2 + 1$ (D) None
14. If $0 < \alpha < \beta < \gamma < \frac{\pi}{2}$, then the equation $\frac{1}{x - \sin \alpha} + \frac{1}{x - \sin \beta} + \frac{1}{x - \sin \gamma} = 0$ has :
- (A) Imaginary roots (B) Real and equal roots
(C) Real and unequal roots (D) Rational roots
15. If α, β be the roots of the equation $(x-a)(x-b)+c=0$ ($c \neq 0$), then the roots of the equation $(x-c-\alpha)(x-c-\beta)=c$ are :
- (A) a and $b+c$ (B) $a+c$ and b (C) $a+c$ and $b+c$ (D) a and b
16. The values of x satisfying $|x^2 + 4x + 3| + (2x + 5) = 0$ are:
- (A) $-4, -1 - \sqrt{3}$ (B) $4, 1 + \sqrt{3}$ (C) $-4, 1 - \sqrt{3}$ (D) $-4, 1 + \sqrt{3}$
17. Let α, β be the roots of $x^2 + bx + 1 = 0$. Then the equation whose roots are $-\left(\alpha + \frac{1}{\beta}\right)$ and $-\left(\beta + \frac{1}{\alpha}\right)$ is :
- (A) $x^2 = 0$ (B) $x^2 + 2bx + 4 = 0$ (C) $x^2 - 2bx + 4 = 0$ (D) $x^2 - bx + 1 = 0$
18. If $x = \sqrt{7 + 4\sqrt{3}}$, $x + \frac{1}{x} =$
- (A) 4 (B) 6 (C) 3 (D) 2
19. What is the solution set of the inequality $\frac{1}{|x|-3} < \frac{1}{2}$?
- (A) $x \in (-\infty, -3) \cup (3, \infty)$ (B) $x \in (-\infty, -5) \cup (-3, 3) \cup (5, \infty)$
(C) $x \in (-\infty, -5) \cup (3, \infty)$ (D) $x \in (-\infty, -3) \cup (5, \infty)$
20. What is the solution set of the inequality $\left| \frac{x^2 - 5x + 4}{x^2 - 4} \right| \leq 1$?
- (A) $x \in [0, 2) \cup \left[\frac{5}{2}, \infty\right)$ (B) $x \in \left[0, \frac{8}{5}\right) \cup \left[\frac{5}{2}, \infty\right)$
(C) $x \in \left[0, \frac{8}{5}\right] \cup \left[\frac{5}{2}, \infty\right)$ (D) None of these

21. If the difference of the roots of the equation $x^2 + px + q = 0$ be unity, then $(p^2 + 4q^2)$ is equal to :
 (A) $(1+2q)^2$ (B) $(1-2q)^2$ (C) $4(p-q)^2$ (D) $2(p-q)^2$
22. Solution set of the equation $3^{2x^2} - 2 \cdot 3^{x^2+x+6} + 3^{2(x+6)} = 0$ is :
 (A) $\{-3, 2\}$ (B) $\{6, -1\}$ (C) $\{-2, 3\}$ (D) $\{1, -6\}$
23. If α, β, γ are the roots of the quadratic equation $x^3 - px^2 + qx - r = 0$, then the value of $\sum \alpha^2 \beta$ is equal to :
 (A) $pq + 3r$ (B) $pq + r$ (C) $pq - 3r$ (D) $\frac{q^2}{r}$
24. If $y = ax^2 + bx + c$ represents the curve given in the figure and $b^2 = 2(b + 2ac)$, where $a \neq 0$ and $AP = 3$ units, then $OP =$
 (A) $\frac{3}{2}$ (B) $\frac{3}{4}$
 (C) 3 (D) 6
- 
25. The coefficient of x in a quadratic equation $x^2 + px + q = 0$ was taken as 17 in place of 13 and its roots were found to be -2 and -15 . The roots of the original equation are :
 (A) $2, 15$ (B) $10, 3$ (C) $-10, -3$ (D) $-2, -15$
26. If the equation $(\cos p - 1)x^2 + x \cos p + \sin p = 0$ in the variable x , has real roots, then p can take any value in the interval :
 (A) $(0, 2\pi)$ (B) $(-\pi, 0)$ (C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (D) $(0, \pi)$
27. If a, b, c are in A.P. and if $(b - c)x^2 + (c - a)x + (a - b) = 0$ and $2(c + a)x^2 + (b + c)x = 0$ have a common root, then :
 (A) a^2, b^2, c^2 are in A.P. (B) a^2, c^2, b^2 are in A.P.
 (C) a^2, c^2, b^2 are in G.P. (D) None of these
28. If α, β are the roots of $x^2 + px + q = 0$ and also of $x^{2n} + p^n x^n + q^n = 0$ and if $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $x^n + 1 + (x+1)^n = 0$, then n is :
 (A) An odd integer (B) An even integer (C) Any integer (D) None of these
29. If $\log_{10}(x^3 + y^3) - \log_{10}(x^2 + y^2 - xy) \leq 2$, then the maximum value of xy , for all $x \geq 0, y \geq 0$ is :
 (A) 2500 (B) 3000 (C) 1200 (D) 3500
30. The largest interval for which $x^{12} - x^9 + x^4 - x + 1 > 0$ is :
 (A) $-4 < x < 0$ (B) $0 < x < 1$ (C) $-100 < x < 100$ (D) $-\infty < x < \infty$
31. If $\frac{1}{\sqrt{\alpha}}$ and $\frac{1}{\sqrt{\beta}}$ are the roots of equation, $ax^2 + bx + 1 = 0$ ($a \neq 0, a, b \in R$), then the equation, $x(x + b^3) + (a^3 - 3abx) = 0$ has roots :
 (A) $\alpha^{3/2}$ and $\beta^{3/2}$ (B) $\alpha\beta^{1/2}$ and $\alpha^{1/2}\beta$ (C) $\sqrt{\alpha\beta}$ and $\alpha\beta$ (D) $\alpha^{-\frac{3}{2}}$ and $\beta^{-\frac{3}{2}}$

32. The sum of the roots of the equation, $x^2 + |2x - 3| - 4 = 0$, is :
 (A) $-\sqrt{2}$ (B) $\sqrt{2}$ (C) -2 (D) 2
33. The equation $\sqrt{3x^2 + x + 5} = x - 3$, where x is real, has :
 (A) no solution (B) exactly one solution
 (C) exactly two solutions (D) exactly four solutions
34. If non-zero numbers b and c are such that $\min f(x) > \max g(x)$, where $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ ($x \in R$); then $\left| \frac{c}{b} \right|$ lies in the interval :
 (A) $\left(0, \frac{1}{2}\right)$ (B) $\left[\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$ (C) $\left[\frac{1}{\sqrt{2}}, \sqrt{2}\right]$ (D) $(\sqrt{2}, \infty)$
35. The value of 'a' for which one root of the equation $x^2 - (a+1)x + a^2 + a - 8 = 0$ exceeds 2 and the other is lesser than 2, are given by :
 (A) $3 < a < 10$ (B) $a \geq 10$ (C) $-2 < a < 3$ (D) $a \leq -2$
36. If α and β are roots of the equation $x^2 + px + \frac{3p}{4} = 0$, such that $|\alpha - \beta| = \sqrt{10}$, then p belongs to the set :
 (A) $\{2, -5\}$ (B) $\{-3, 2\}$ (C) $\{-2, 5\}$ (D) $\{3, -5\}$
37. If the equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a common root different from -1 , then $|b|$ is equal to :
 (A) $\sqrt{2}$ (B) 2 (C) 3 (D) $\sqrt{3}$
38. If x is a solution of the equation, $\sqrt{2x+1} - \sqrt{2x-1} = 1$, $\left(x \geq \frac{1}{2}\right)$, then $\sqrt{4x^2 - 1}$ is equal to :
 (A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) $2\sqrt{2}$
39. If $2 + 3i$ is one of the roots of the equation $2x^3 - 9x^2 + kx - 13 = 0$, $k \in R$, then the real root of this equation:
 (A) does not exist (B) exists and is equal to $1/2$
 (C) exists and is equal to $-1/2$ (D) exists and is equal to 1
40. If two roots of the equation, $(a-1)(x^4 + x^2 + 1) + (a+1)(x^2 + x + 1)^2 = 0$ are real and distinct, then the set of all values of 'a' is :
 (A) $\left(-\frac{1}{2}, 0\right)$ (B) $(-\infty, -2) \cup (2, \infty)$ (C) $\left(-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{2}\right)$ (D) $\left(0, \frac{1}{2}\right)$
41. If α and β are roots of the equation, $x^2 - 4\sqrt{2}kx + 2e^{4\ln k} - 1 = 0$ for some k , and $\alpha^2 + \beta^2 = 66$, then $\alpha^3 + \beta^3$ is equal to :
 (A) $248\sqrt{2}$ (B) $280\sqrt{2}$ (C) $-32\sqrt{2}$ (D) $-280\sqrt{2}$
42. The complete set of values of a so that the equation $x^2 + (3-2a)x + a = 0$ has exactly 1 root in $(-1, 2)$ is:
 (A) $\left(-\infty, \frac{2}{3}\right) \cup \left(\frac{10}{3}, \infty\right)$ (B) $\left(-\infty, \frac{2}{3}\right] \cup \left(\frac{10}{3}, \infty\right)$
 (C) $\left(-\infty, \frac{2}{3}\right) \cup \left[\frac{10}{3}, \infty\right)$ (D) $\left(-\infty, \frac{2}{3}\right] \cup \left[\frac{10}{3}, \infty\right)$

43. The equation $x^2 + bx + c = 0$ has distinct roots. If 2 is subtracted from each root the result are the reciprocal of the original roots, then the value of $b^2 + c^2$ equals:
 (A) 2 (B) 3 (C) 4 (D) 5
44. If α, β are the roots of the quadratic equation $x^2 + px + q = 0$ and γ, δ are the roots of $x^2 + px - r = 0$ then $(\alpha - \gamma)(\alpha - \delta)$ is equal to:
 (A) $q + r$ (B) $q - r$ (C) $-(q + r)$ (D) $-(p + q + r)$
45. Minimum value of $y = x^2 - 3x + 5, x \in [-4, 1]$ is a . Then the value of $[a]$ is; $[.]$ denotes greatest integer function
 (A) 3 (B) $\frac{11}{4}$ (C) 0 (D) 9
46. Let $a, b, c \in R$ be such that $a + b + c < 0, a - b + c < 0$ and $c > 0$. If ' α ' and ' β ' are roots of the equation $ax^2 + bx + c = 0$, then value of $[\alpha] + [\beta]$ is : (where $[.]$ denotes greatest integer of x)
 (A) 2 (B) 1 (C) -1 (D) 0
47. If $a \neq 0$ and the equation $ax^2 + bx + c = 0$ has two roots α & β such that $\alpha < -3$ and $\beta > 2$ which of the following is always true?
 (A) $a(a + |b| + c) > 0$ (B) $a(a + |b| + c) < 0$
 (C) $9a - 3b + c > 0$ (D) $(9a - 3b + c)(4a + 2b + c) < 0$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

48. The least integral value of m for which every solution of the inequality $1 \leq x \leq 2$ is a solution of the inequality $x^2 - mx + 1 < 0$
49. Let $f(x) = (a^2 + a + 2)x^2 - (a + 4)x - 7, x \in R$. If unity lies between the roots of equation $f(x) = 0$ then the number of integral values of a is:
50. The smallest positive integer P for which the expression $x^2 - 2px + 3p + 4$ is negative for at least one real x is:
51. The sum of all integral values of λ for which $(\lambda^2 + \lambda - 2)x^2 + (\lambda + 2)x < 1 \quad \forall x \in R$, is k . Then the value of $|k|$ is_____.
52. If exactly one root of the quadratic equation $x^2 - (a + 1)x + 2a = 0$ lies in the interval $(0, 3)$, then the smallest positive integer value of a is
53. The least integral value of ' k ' for which $(k - 1)x^2 - (k + 1)x + (k + 1)$ is positive for all real value of x is:
54. If α and β are the roots of the quadratic equation $4x^2 + 2x - 1 = 0$, then the value of $\sum_{r=1}^{\infty} (\alpha^r + \beta^r)$ is:

55. If the equation $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ and $x^2 + (a+b)x + 36 = 0$ have a common positive root, then $b - 2a$ is equal to_____.
56. If α, β are the root of the equation $x^2 - x + 1 = 0$ then the value of $\left| \alpha^{2020} + \beta^{2020} \right|$ is _____.
57. Let α, β be the roots of $x^2 - 4x + A = 0$ and γ, δ be the roots of $x^2 - 36x + B = 0$. If $\alpha, \beta, \gamma, \delta$ form an increasing G.P. and $A^t = B$ then the value of t equals:
58. If $3x^2 - 17x + 10 = 0$ and $x^2 - 5x + m = 0$ has a common root, then the sum of all possible real values of 'm' is A then the value of $[A]$ is. $[.]$ = greatest integer function
59. If r and s are variable satisfying the equation $\frac{1}{r+s} = \frac{1}{r} + \frac{1}{s}$. Then the value of $\left(\frac{r}{s}\right)^3$ is:
60. Let 'k' be a real number. The minimum number of distinct real roots possible of the equation $(3x^2 + kx + 3)(x^2 + kx - 1) = 0$ is:
61. If α, β be the roots of $4x^2 - 16x + \lambda = 0, \lambda \in R$, such that $1 < \alpha < 2$ and $2 < \beta < 3$ then the number of integral solution of λ is :
62. If equation and $2x^2 + 3x + 4 = 0$ and $ax^2 + bx + c = 0$ have a common root and $a, b, c \in N$. Then the minimum value of $a + b + c$ is:

Straight Line

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED ‘*’ MAY HAVE MORE THAN ONE CORRECT OPTION.

1. Given the triangle with vertices $A(10, 4)$, $B(-4, 9)$, $C(-2, 1)$, the equation of median through B is :
(A) $3x + 4y = 92$ **(B)** $13x + 16y = 92$ **(C)** $13x - 61y = 9$ **(D)** $3x - 4y = 9$
- *2. The points on $x + y = 4$ that lie at a distance 1 unit from the line $4x + 3y - 10 = 0$ are :
(A) $(-3, -1)$ **(B)** $(-7, 11)$ **(C)** $(3, 1)$ **(D)** $(7, -11)$
3. The distance of the point $(2, 3)$ from the line $2x - 3y + 9 = 0$ measured along a line $x - y + 1 = 0$ is :
(A) $4\sqrt{2}$ **(B)** $2\sqrt{2}$ **(C)** $\sqrt{2}$ **(D)** $\frac{1}{\sqrt{2}}$
4. The lines $ax + by + c = 0$, $bx + cy + a = 0$ and $cx + ay + b = 0$ ($a \neq b \neq c$) are concurrent, if :
(A) $a^3 + b^3 + c^3 + 3abc = 0$ **(B)** $a^2 + b^2 + c^2 - 3abc = 0$
(C) $a + b + c = 0$ **(D)** None of these
5. In what direction should a line be drawn through the point $(1, 2)$ so that its point of intersection with the line $x + y = 4$ is at a distance of $\frac{\sqrt{6}}{3}$ from the given point ?
(A) 75° **(B)** 60° **(C)** 90° **(D)** 45°
6. The values of θ and p , if the equation $x \cos \theta + y \sin \theta = p$ is the normal form of the line $\sqrt{3}x + y + 2 = 0$ are:
(A) $210^\circ, 1$ **(B)** $210^\circ, 2$ **(C)** $220^\circ, 3$ **(D)** None of these
7. P is a point on either of the two lines $y - \sqrt{3}|x| = 2$ at a distance 5 units from their point of intersection. The coordinate of the foot of the perpendicular from P on the bisector of the angle between them are :
(A) $\left(0, \frac{4+5\sqrt{3}}{2}\right)$ or $\left(0, \frac{4-5\sqrt{3}}{2}\right)$ depending on which the point P is taken
(B) $\left(0, \frac{4+5\sqrt{3}}{2}\right)$ **(C)** $\left(0, \frac{4-5\sqrt{3}}{2}\right)$ **(D)** $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$
8. A square of area 25 sq units is formed by taking two sides as $3x + 4y = k_1$ and $3x + 4y = k_2$, then $|k_1 - k_2|$ is:
(A) 5 units **(B)** 1 unit **(C)** 25 unit **(D)** None of these
9. If t_1 and t_2 are roots of the equation $t^2 + \lambda t + 1 = 0$, where λ is an arbitrary constant. Then, the line joining the points $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ always passes through a fixed point whose coordinates are :
(A) $(a, 0)$ **(B)** $(-a, 0)$ **(C)** $(0, a)$ **(D)** $(0, -a)$
10. The straight line $ax + by + c = 0$ where $abc \neq 0$ will pass through the first quadrant if :
(A) $ac > 0, bc > 0$ **(B)** $c > 0$ & $bc < 0$ **(C)** $bc > 0$ & $ac > 0$ **(D)** $ac < 0$ & $bc < 0$
11. A light ray coming along the line $3x + 4y = 5$ gets reflected from the line $ax + by = 1$ and goes along the line $5x - 12y = 10$. Then,
(A) $a = \frac{64}{115}, b = \frac{112}{15}$ **(B)** $a = \frac{14}{15}, b = -\frac{8}{115}$ **(C)** $a = \frac{64}{115}, b = -\frac{8}{115}$ **(D)** $a = \frac{64}{15}, b = \frac{14}{15}$

12. The lines $x + y = |a|$ and $ax - y = 1$ intersect each other in the first quadrant. Then, the set of all possible values of a in the interval :
- (A) $(-1, 1]$ (B) $(0, \infty)$ (C) $[1, \infty)$ (D) $(-1, \infty)$
13. If (a, a^2) falls inside the angle made by the lines, $y = \frac{x}{2}, x > 0$ and $y = 3x, x > 0$, then $a \in$:
- (A) $\left(\frac{1}{2}, 3\right)$ (B) $\left(-3, -\frac{1}{2}\right)$ (C) $\left(0, \frac{1}{2}\right)$ (D) $(3, \infty)$
14. Through the point $P(\alpha, \beta)$, where $\alpha\beta > 0$ the straight line $\frac{x}{\alpha} + \frac{y}{\beta} = 1$ is drawn so as to form with coordinate axes a triangle of area S . If $ab > 0$, then the least value of S is :
- (A) $\alpha\beta$ (B) $2\alpha\beta$ (C) $4\alpha\beta$ (D) None of these
15. If the equal sides AB and AC (each equal to a) of a right angled isosceles $\triangle ABC$ be produced to P and Q so that $BP \cdot CQ = AB^2$, then the line PQ always passes through the fixed point :
- (A) $(a, 0)$ (B) $(0, a)$ (C) (a, a) (D) None of these
16. If $u = a_1x + b_1y + c_1 = 0$ and $v = a_2x + b_2y + c_2 = 0$ and $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} = \frac{c_1}{c_2}$, then $u + kv = 0$ represents :
- (A) $u = 0$ (B) A family of concurrent lines
(C) A family of parallel line (D) None of the above
17. A variable line is drawn through the origin O . Two points A and B are taken on the line such that $OA = 1$ and $OB = 2$ unit. Through points A and B two lines are drawn making equal angle α with the line AB . Then the locus of the point of intersection of the lines, is :
- (A) $x^2 + y^2 = \frac{9 + \tan^2 \alpha}{4}$ (B) $x^2 + y^2 = \frac{9 - \tan^2 \alpha}{4}$
(C) $x^2 + y^2 = \frac{9 + \tan^2 \alpha}{2}$ (D) $x^2 + y^2 = \frac{9 + 2 \tan^2 \alpha}{4}$
18. The point $(4, 1)$ undergoes the following three transformation successively.
- I. Reflection about the line $y = x$
II. Transformation through a distance 2 unit along the positive direction of X -axis.
III. Rotation through an angle of $\pi/4$ about the origin in the anti-clockwise direction.
The final position of the point is given by the coordinates.
- (A) $\left(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ (B) $(-2, 7\sqrt{2})$ (C) $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ (D) $(\sqrt{2}, 7\sqrt{2})$
19. If the line $y = \tan \theta x$ cut the curve $x^3 + xy^2 + 2x^2 + 2y^2 + 3x + 1 = 0$ at the points A, B and C . If OA, OB, OC are in $H.P.$, then $\tan \theta$ is equal to :
- (A) ± 1 (B) 0 (C) 2 (D) -2
20. The number of points on the line $x + y = 4$ which are unit distance apart from the line $2x + 2y = 5$ is :
- (A) 0 (B) 1 (C) 2 (D) ∞
21. The sides of a quadrilateral are given by $xy(x-2)(y-3) = 0$. The equation of the line parallel to $x - 4y = 0$, which divides the quadrilateral into two equal regions, is :
- (A) $x - 4y - 1 = 0$ (B) $x - 4y + 5 = 0$ (C) $x - 4y + 1 = 0$ (D) $x - 4y + 3 = 0$

22. The circumcentre of the triangle formed by the lines $xy + 2x + 2y + 4 = 0$ and $x + y + 2 = 0$ is :
(A) $(0, 0)$ **(B)** $(-1, -1)$ **(C)** $(-1, -2)$ **(D)** $(-2, -2)$
23. The equation $(1+2k)x + (1-k)y + k = 0$, k being parameter represents a family of lines. The line which belongs to this family and is at a maximum distance from the point $(1, 4)$ is :
(A) $33x + 12y + 7 = 0$ **(B)** $12x + 33y - 7 = 0$
(C) $4x - y + 7 = 0$ **(D)** $12x - 33y + 7 = 0$
24. The diagonal of the rectangle formed by the lines $x^2 - 7x + 6 = 0$ and $y^2 - 14y + 40 = 0$ is :
(A) $5x + 6y = 0$ **(B)** $5x - 6y = 0$ **(C)** $6x - 5y + 14 = 0$ **(D)** $6x - 5y - 14 = 0$
25. A man at the crossing of two roads $x - 2y - 4 = 0$ and $2x - y - 4 = 0$, starts walking along the bisector of the acute angle between the roads and after covering a distance 2 km reaches the bank of a straight river at right angle to its path. Then the co-ordinate(s) of the point where his path meets the bank is(are) :
(A) $\left(\frac{4}{3} + \sqrt{2}, \frac{4}{3} - \sqrt{2}\right)$ **(B)** $\left(\frac{4}{3} \pm \sqrt{2}, -\frac{4}{3} \pm \sqrt{2}\right)$
(C) $\left(\frac{4}{3} - 2\sqrt{2}, \frac{4}{3} - \sqrt{2}\right)$ **(D)** $\left(\frac{4}{3} + 3\sqrt{2}, \frac{4}{3} - \sqrt{2}\right)$
26. If $f(x+y) = f(x)f(y) \forall x, y \in R$ and $f(1) = 2$ then area enclosed by $3|x| + 2|y| \leq 8$ is :
(A) $f(4)$ square units **(B)** $\frac{1}{2} f(6)$ square units
(C) $\frac{1}{3} f(6)$ square units **(D)** $\frac{1}{3} f(5)$ square units
27. $ABCD$ is square whose vertices A, B, C and D are $(0, 0), (2, 0), (2, 2)$ and $(0, 2)$ respectively. This square is rotated in the xy plane with an angle of 30° in anticlockwise direction about an axis passing through the vertex A , then the equation of the diagonal BD of this rotated square is :
(A) $\sqrt{3}x + (1 - \sqrt{3})y = \sqrt{3}$ **(B)** $(1 + \sqrt{3})x - (1 - \sqrt{2})y = 2$
(C) $(2 - \sqrt{3})x + y = 2(\sqrt{3} - 1)$ **(D)** None of these
28. The Cartesian co-ordinates (x, y) of a point on a curve are given by $x : y : 1 = t^3 : t^2 - 3 : t - 1$ where t is a parameter, then the points given by $t = a, b, c$ are collinear, if :
(A) $abc + 3(a + b + c) = ab + bc + ca$ **(B)** $3abc + 2(a + b + c) = ab + bc + ca$
(C) $abc + 2(a + b + c) = 3(ab + bc + ca)$ **(D)** None of these
29. If $A\left(\frac{\sin \alpha}{3} - 1, \frac{\cos \alpha}{2} - 1\right)$ and $B(1, 1)$, $\alpha \in [-\pi, \pi]$ are two points on the same side of the line $3x - 2y + 1 = 0$, then α belongs to the interval :
(A) $\left[-\pi, -\frac{3\pi}{4}\right) \cup \left(\frac{\pi}{4}, \pi\right]$ **(B)** $[-\pi, \pi]$ **(C)** ϕ **(D)** None of these

30. If p_1, p_2, p_3 be the length of perpendiculars from the points $(m^2, 2m), (mm', m+m')$ and $(m'^2, 2m')$ respectively on the line $x \cos \alpha + y \sin \alpha + \frac{\sin^2 \alpha}{\cos \alpha} = 0$, then p_1, p_2, p_3 are in :
 (A) A.P. (B) G.P. (C) H.P. (D) None of these
31. The vertices of a triangle are $A(x_1, x_1 \tan \alpha)$, $B(x_2, x_2 \tan \beta)$ and $C(x_3, x_3 \tan \gamma)$. If the circumcentre of ΔABC coincides with the origin and $H(a, b)$ be its orthocentre, then a/b is equal to :
 (A) $\frac{\cos \alpha + \cos \beta + \cos \gamma}{\cos \alpha \cos \beta \cos \gamma}$ (B) $\frac{\sin \alpha + \sin \beta + \sin \gamma}{\sin \alpha \sin \beta \sin \gamma}$
 (C) $\frac{\tan \alpha + \tan \beta + \tan \gamma}{\tan \alpha \tan \beta \tan \gamma}$ (D) $\frac{\cos \alpha + \cos \beta + \cos \gamma}{\sin \alpha + \sin \beta + \sin \gamma}$
32. The equation $\sqrt{(x-2)^2 + y^2} + \sqrt{(x+2)^2 + y^2} = 4$ represents :
 (A) a circle (B) a line (C) a parabola (D) an ellipse
33. If the point $(\cos \theta, \sin \theta)$ does not fall in that angle between the lines $y = |x-1|$ in which the origin lies, then θ belongs to :
 (A) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ (B) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (C) $(0, \pi)$ (D) None of these
34. The point $(\alpha^2, \alpha+1)$ lies in the angle between the lines $3x-4y+1=0$ and $x+2y-5=0$ containing the origin, then $[\alpha]$ can be (where $[.]$ denotes the greater integer function) :
 (A) -1 (B) 5 (C) 6 (D) 4
35. A variable straight line through the point of intersection of the lines $\frac{x}{3} + \frac{y}{4} = 1$ and $\frac{x}{4} + \frac{y}{3} = 1$ meets the coordinate axes in A and B. If locus of the mid point of AB is $2xy = k(x+y)$, then the value of $343k$ must be :
 (A) 855 (B) 588 (C) 577 (D) 686
36. If m_1 and m_2 are the roots of the equation $x^2 + (\sqrt{3}+2)x + (\sqrt{3}-1) = 0$ and if area of the triangle formed by the lines $y = m_1x$, $y = m_2x$, and $y = c$ is $(a+b)c^2$, then the value of $2008(a^2 + b^2)$ must be :
 (A) 2255 (B) 5522 (C) 3344 (D) 4433
37. If all the 3 vertices of an isosceles right angle triangle be integral points and length of base is also an integer, then which of the point is never a rational point (A point $P(x, y)$ is integer point if both x and y are integers and point is rational, if both x and y are rational) :
 (A) Centroid (B) Incentre (C) Circumcentre (D) Orthocentre
38. The point $(2, 1)$ is translated parallel to the line $L : x - y = 4$ by $2\sqrt{3}$ units. If the new point Q lies in the third quadrant, then the equation of the passing through Q and perpendicular to L is :
 (A) $x + y = 2 - \sqrt{6}$ (B) $x + y = 3 - 3\sqrt{6}$ (C) $x + y = 3 - 2\sqrt{6}$ (D) $2x + 2y = 1 - \sqrt{6}$
39. A ray light is incident along a line which meets another line, $7x - y + 1 = 0$, at the point $(0, 1)$. the ray is then reflected from this point along the line, $y + 2x = 1$. Then the equation of the line of incidence of the ray of light is :
 (A) $41x - 38y + 38 = 0$ (B) $41x + 25y - 25 = 0$
 (C) $41x + 38y - 38 = 0$ (D) $41x - 25y + 25 = 0$

40. A straight line through origin O meets the lines $3y = 10 - 4x$ and $8x + 6y + 5 = 0$ at points A and B respectively. Then O divides the segment AB in the ratio :
 (A) 2 : 3 (B) 1 : 2 (C) 4 : 1 (D) 3 : 4
41. The points $\left(0, \frac{8}{3}\right)$, $(1, 3)$ and $(82, 30)$:
 (A) form an obtuse angled triangle (B) form an acute angled triangle
 (C) form a right angled triangle (D) lie on a straight line
42. Let L be the line passing through the point $P(1, 2)$ such that its intercepted segment between the co-ordinate axes is bisected at P . If L_1 is the line perpendicular to L and passing through the point $(-2, 1)$, then the point of intersection of L and L_1 is :
 (A) $\left(\frac{4}{5}, \frac{12}{5}\right)$ (B) $\left(\frac{11}{20}, \frac{29}{10}\right)$ (C) $\left(\frac{3}{10}, \frac{17}{5}\right)$ (D) $\left(\frac{3}{5}, \frac{23}{10}\right)$
43. A straight line L through the point $(3, -2)$ is inclined at an angle of 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x -axis, then the equation of L is :
 (A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$ (B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$
 (C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$ (D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$
44. If the incentre of an equilateral triangle is $(1, 1)$ and the equation of its one side is $3x + 4y + 3 = 0$, then the equation of the circumcircle of this triangle is :
 (A) $x^2 + y^2 - 2x - 2y - 2 = 0$ (B) $x^2 + y^2 - 2x - 2y - 14 = 0$
 (C) $x^2 + y^2 - 2x - 2y + 2 = 0$ (D) $x^2 + y^2 - 2x - 2y - 7 = 0$
45. The base of an equilateral triangle is along the line given by $3x + 4y = 9$. If a vertex of the triangle is $(1, 2)$, then the length of a side of the triangle is :
 (A) $\frac{2\sqrt{3}}{15}$ (B) $\frac{4\sqrt{3}}{15}$ (C) $\frac{4\sqrt{3}}{5}$ (D) $\frac{2\sqrt{3}}{5}$
46. If a line intercepted between the coordinate axes is trisected at a point $A(4, 3)$, which is nearer to x -axis, then its equation is :
 (A) $4x - 3y = 4$ (B) $x + 3y = 13$ (C) $3x + 2y = 18$ (D) $3x + 8y = 36$
47. The circumcentre of a triangle lies at the origin its centroid is the mid point of the line segment joining the points $(a^2 + 1, a^2 + 1)$ and $(2a, -2a)$, $a \neq 0$. Then for any a , the orthocentre of this triangle lies on the line:
 (A) $y - 2ax = 0$ (B) $y - (a^2 + 1)x = 0$
 (C) $y + x = 0$ (D) $(a - 1)^2 x - (a + 1)^2 y = 0$
48. If a line L is perpendicular to the line $5x - y = 1$, and the area of the triangle formed by the line L and the coordinate axes is 5, then the distance of line L from the line $x + 5y = 0$ is :
 (A) $\frac{7}{\sqrt{5}}$ (B) $\frac{5}{\sqrt{13}}$ (C) $\frac{7}{\sqrt{13}}$ (D) $\frac{5}{\sqrt{7}}$

49. A light ray emerging from the point source placed at $P(1, 3)$ is reflected at a point Q in the axis of x . If the reflected ray passes through the point $R(6, 7)$, then the abscissa of Q is :
 (A) 1 (B) 3 (C) $7/2$ (D) $5/2$
50. If the three lines $x - 3y = p$, $ax + 2y = q$ and $ax + y = r$ form a right-angled triangle then :
 (A) $a^2 - 9a + 18 = 0$ (B) $a^2 - 6a - 12 = 0$ (C) $a^2 - 6a - 18 = 0$ (D) $a^2 - 9a + 12 = 0$
51. If the x -intercept of some line L is double as that of the line, $3x + 4y = 12$ and the y -intercept of L is half as that of the same line, then the slope of L is :
 (A) -3 (B) $-3/8$ (C) $-3/2$ (D) $-3/16$
52. If the extremities of the base of an isosceles triangle are the points $(2a, 0)$ and $(0, a)$ and the equation of one of the sides is $x = 2a$, then the area of the triangle, in square units is :
 (A) $\frac{5}{4}a^2$ (B) $\frac{5}{2}a^2$ (C) $\frac{25a^2}{4}$ (D) $5a^2$
53. Let θ_1 be the angle between two lines $2x + 3y + c_1 = 0$ and $-x + 5y + c_2 = 0$, and θ_2 be the angle between two lines $2x + 3y + c_1 = 0$ and $-x + 5y + c_3 = 0$, where c_1, c_2, c_3 are any real numbers :
Statement 1 : If c_2 and c_3 are proportional, then $\theta_1 = \theta_2$.
Statement 2 : $\theta_1 = \theta_2$ for all c_2 and c_3 .
 (A) Statement 1 is true; Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
 (B) Statement 1 is true; Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
 (C) Statement 1 is true; Statement 2 is false.
 (D) Statement 1 is false; Statement 2 is true.

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

54. If n_1 is the number of points on the line $3x + 4y = 5$ which is at distance of $1 + \sin^2 \theta$ units from $(2, 3)$ and n_2 denotes the number of points on the line $3x + 4y = 5$ which is at distance of $\sec^2 \theta + 2 \operatorname{cosec}^2 \theta$ units from $(1, 3)$, then find the sum of roots of equations $n_2 x^2 - 6x + n_1 = 0$.
55. The slopes of three sides of a triangle ABC are -1, -2, 3 respectively. If the orthocenter of triangle ABC is origin, then the locus of its centroid is $y = \frac{a}{b}x$ where a, b are relatively prime then $b - a$ is equal to.
56. Consider two lines $L_1 \equiv x - y = 0$ and $L_2 \equiv x + y = 0$ and a moving point $P(x, y)$. Let $d(P, L_i), i = 1, 2$ represents the perpendicular distance of the point P from L_i . If point P moves in certain region R in such a way that $\sum_{i=1}^2 d(P, L_i) \in [2, 4]$. Let the area of region R is A, then find $\frac{A}{4}$.

57. A variable line L_1 cuts $y = 3x + 1$ and $y = -2x + 3$ at points P_1 and P_2 . If the locus of midpoints of P_1 and P_2 is lines L_2 with undefined slope where slope of L_1 is constant. If slope of L_1 is $\frac{p}{q}$, where p, q are coprime natural number, then find $p + q$.
58. Let A, B, C lies on lines $y = x$, $y = 2x$ and $y = 3x$ respectively. Also AB passes through fixed point (1, 0), BC passes through fixed point (0, -1), then AC also passes through fixed point (h, k), find the value of $h + k$.
59. In a $\triangle ABC$, the vertex A is (1, 1) and orthocenter is (2, 4). If the sides AB and BC are members of the family of straight lines $ax + by + c = 0$. Where a, b, c are in A.P. then the coordinates of vertex C are (h, k). Find the value of $2h + 12k$.
60. Let P be any point on the line $x - y + 3 = 0$ and A be a fixed point (3, 4). If the family of lines given by the equation $(3\sec\theta + 5\operatorname{cosec}\theta)x + (7\sec\theta - 3\operatorname{cosec}\theta)y + 11(\sec\theta - \operatorname{cosec}\theta) = 0$ are concurrent at a point B for all permissible values of θ and maximum value of $|PA - PB| = 2\sqrt{2n}$ ($n \in \mathbb{N}$), then find the value of n .
61. The base of an isosceles triangle is the intercept made by the line $x + 2y = 4$ with the coordinate axes. If the equations of the equal sides be $x = 4$ and $y = mx + c$ then find the value of $8m + c$.
62. The equation of a line through the mid point of the sides AB and AD of rhombus ABCD, whose one diagonal is $3x - 4y + 5 = 0$ and one vertex is A(3, 1) is $ax + by + c = 0$. Find the absolute value of $(a + b + c)$ where a, b, c are integers expressed in lowest form.
63. In a $\triangle ABC$, $A \equiv (\alpha, \beta)$, $B(1, 2)$, $C(2, 3)$ and point A lies on line $y = 2x + 3$, where $\alpha, \beta \in I$. If the area of $\triangle ABC$ be such that area of triangle lies in interval [2, 3). Find the number of all possible coordinates of A.
64. Consider $\triangle ABC$ with $A(m - m - 1)$, $B(-1, 0)$, $C(l, l + 1)$ is such that a line of slope 2, drawn through centroid of $\triangle ABC$ meets the circumcentre of $\triangle ABC$ on y-axis, then find the value of $l + m$.
65. Find the area of the triangle formed by the line $x + y = 3$ and angle bisector of the pair of straight lines $x^2 - y^2 + 2y - 1 = 0$
66. Find the area enclosed by the graphs of $|x + y| = 2$ and $|x| = 1$ is ____.
67. A piece of cheese is located at (12, 10) in a coordinate plane. A mouse is at (4, -2) and is running up the line $y = -5x + 18$. At the point (a, b) the mouse starts getting farther from the cheese rather than closer to it. The value of $(a + b)$ is
68. The line $x = c$ cuts the triangle with corners (0, 0), (1, 1) and (9, 1) into two regions. For the area of the two regions to be the same c must be equal to:

Circles

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT.

- The equation of a circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length $3a$ is :
 (A) $x^2 + y^2 = 9a^2$ (B) $x^2 + y^2 = 16a^2$ (C) $x^2 + y^2 = 4a^2$ (D) $x^2 + y^2 = a^2$
- The circles $x^2 + y^2 + 2g_1x - a^2 = 0$ and $x^2 + y^2 + 2g_2x - a^2 = 0$ cut each other orthogonally. If p_1 and p_2 are perpendicular from $(0, a)$ and $(0, -a)$ on a common tangent of these circles, then $p_1 p_2$ is equal to :
 (A) $a^2/2$ (B) a^2 (C) $2a^2$ (D) $a^2 - 2$
- A rhombus is inscribed in the region common to the two circles $x^2 + y^2 - 4x - 12 = 0$ and $x^2 + y^2 + 4x - 12 = 0$ with two of its vertices on the line joining the centres of the circles. The area of the rhombus is :
 (A) $8\sqrt{3}$ sq. units (B) $4\sqrt{3}$ sq. units (C) $6\sqrt{3}$ sq. units (D) None of these
- Two circles with radii a and b touch each other externally such that θ is the angle between the direct common tangents ($a < b \leq 2$), then :
 (A) $\theta = 2\cos^{-1}\left(\frac{a-b}{a+b}\right)$ (B) $\theta = 2\tan^{-1}\left(\frac{a+b}{a-b}\right)$ (C) $\theta = 2\sin^{-1}\left(\frac{a+b}{a-b}\right)$ (D) $\theta = 2\sin^{-1}\left(\frac{a-b}{a+b}\right)$
- If the area of the quadrilateral formed by the tangent from the origin to the circle $x^2 + y^2 + 6x - 10y + c = 0$ and the pair of radii at the points of contact of these tangents to the circle is 8 sq. units, then c is a root of the equation :
 (A) $c^2 - 32c + 64 = 0$ (B) $c^2 - 34c + 64 = 0$ (C) $c^2 + 2c - 64 = 0$ (D) $c^2 + 34c - 64 = 0$
- Any chord of the circle $x^2 + y^2 = 25$ subtends a right angle at the centre. Then, the locus of the centroid of the triangle made by the chord and a moving point P on the circle is :
 (A) Parabola (B) Circle (C) Rectangular hyperbola (D) Ellipse
- Find the equation of the circle passing through $(1, 0)$ and $(0, 1)$ and having the smallest possible radius :
 (A) $x^2 + y^2 + x + y = 0$ (B) $x^2 + y^2 - x - y = 0$
 (C) $x^2 + y^2 + 2x + 2y = 0$ (D) $x^2 + y^2 - 2x - 2y = 0$
- The circles $(x-a)^2 + (x-b)^2 = c^2$ and $(x-b)^2 + (y-a)^2 = c^2$ touch each other, then :
 (A) $a = b \pm 2c$ (B) $a = b \pm \sqrt{2}c$ (C) $a = b \pm c$ (D) $a = b^2 - c$
- If $4\ell^2 - 5m^2 + 6\ell + 1 = 0$ and the line $\ell x + my + 1 = 0$, touches a fixed circle, then :
 (A) The centre of the circle is at the point $(4, 0)$ (B) The radius of the circle is equal to $\sqrt{5}$
 (C) The circle passes through origin (D) None of these
- If $(1 + \alpha x)^n = 1 + 8x + 24x^2 + \dots$ and a line through $P(\alpha, n)$ cuts the circle $x^2 + y^2 = 4$ in A and B, then $PA \cdot PB$ is equal to :
 (A) 4 (B) 8 (C) 16 (D) 32

11. One of the diameter of the circle circumscribing the rectangle $ABCD$ is $4y = x + 7$. If A and B are the points $(-3, 4)$ and $(5, 4)$ respectively, then the area of rectangle is :
 (A) 16 sq. unit (B) 24 sq. unit (C) 32 sq. unit (D) 48 sq. unit
12. If the circles $x^2 + y^2 + 2ax + c = 0$ and $x^2 + y^2 + 2by + c = 0$ touch each other then :
 (A) $a^{-2} + b^{-2} = c^{-1}$ (B) $a^{-2} + b^{-2} = c^{-2}$ (C) $a + b = 2c$ (D) $\frac{1}{a} + \frac{1}{b} = \frac{2}{c}$
13. The range of values of λ for which the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 4\lambda x + 9 = 0$ have two common tangents, is:
 (A) $\lambda \in \left[-\frac{13}{8}, \frac{13}{8}\right]$ (B) $\lambda > \frac{13}{8}$ or $\lambda < -\frac{13}{8}$
 (C) $1 < \lambda < \frac{13}{8}$ (D) None of these
14. The range of values of m for which the line $y = mx + 2$ cuts the circle $x^2 + y^2 = 1$ at distinct or coincident points is:
 (A) $(-\infty, -\sqrt{3}] \cup [\sqrt{3}, +\infty)$ (B) $[-\sqrt{3}, \sqrt{3}]$
 (C) $[\sqrt{3}, +\infty) [\sqrt{3}, +\infty)$ (D) None of these
15. The length of the tangents from any point on the circle $15x^2 + 15y^2 - 48x + 64y = 0$ to the two circles $5x^2 + 5y^2 - 24x + 32y + 75 = 0$, $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ are in the ratio :
 (A) 1 : 2 (B) 2 : 3 (C) 3 : 4 (D) None of these
16. The equation of a circle which cuts off the three circles $x^2 + y^2 - 3x - 6y + 14 = 0$; $x^2 + y^2 - x - 4y + 8 = 0$ and $x^2 + y^2 + 2x - 6y + 9 = 0$ Orthogonally is:
 (A) $x^2 + y^2 - 2x - 4y + 1 = 0$ (B) $x^2 + y^2 + 2x + 4y + 1 = 0$
 (C) $x^2 + y^2 - 2x + 4y + 1 = 0$ (D) $x^2 + y^2 - 2x - 4y - 1 = 0$
17. The triangle PQR is inscribed in the circle $x^2 + y^2 = 25$. If Q and R have co-ordinates $(3, 4)$ and $(-4, 3)$ respectively, then $\angle QPR$ is equal to :
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$
18. AB is diameter of a circle and C is any point on the circumference of the circle. Then :
 (A) The area of ΔABC is maximum when it is isosceles
 (B) The area of ΔABC is minimum when it is isosceles
 (C) The perimeter of ΔABC is minimum when it is isosceles
 (D) None of these
19. A circle passes through the points $A(1, 0)$, $B(5, 0)$ and touches the Y axis at $C(0, h)$. If $\angle ACB$ is maximum, then :
 (A) $h = \sqrt{5}$ (B) $h = 2\sqrt{5}$ (C) $h = \sqrt{10}$ (D) $h = 2\sqrt{10}$
20. If one of the diameter of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord to the circle with centre $(2, 1)$, then the radius of the circle is :
 (A) $\sqrt{3}$ (B) $\sqrt{2}$ (C) 3 (D) 2

21. If A and B are two given points in a plane such that $\frac{PA}{PB} = k$ (constant) for all P on a given circle, then the value of k can't be equal to :
 (A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2
22. Tangents PA and PB are drawn to $x^2 + y^2 = 4$ from the point $P(3, 0)$. Area of triangle PAB is equal to :
 (A) $\frac{5}{9}\sqrt{5}$ sq. units (B) $\frac{1}{3}\sqrt{5}$ sq. units (C) $\frac{10}{9}\sqrt{5}$ sq. units (D) $\frac{20}{3}\sqrt{5}$ sq. units
23. If the equation of a given circle is $x^2 + y^2 = 36$, then the length of the chord which lies along the line $3x + 4y - 15 = 0$ is :
 (A) $3\sqrt{6}$ (B) $2\sqrt{3}$ (C) $6\sqrt{3}$ (D) None of these
24. The circles $x^2 + y^2 + 2x - 2y + 1 = 0$ and $x^2 + y^2 - 2x - 2y + 1 = 0$ touch each other :
 (A) Externally at $(0, 1)$ (B) Internally at $(0, 1)$
 (C) Externally at $(1, 0)$ (D) Internally at $(1, 0)$
25. The locus of the mid points of the chords of the circle $x^2 + y^2 = 4$ which subtend a right angle at the origin is :
 (A) $x^2 + y^2 = 1$ (B) $x^2 + y^2 = 2$ (C) $x + y = 1$ (D) $x + y = 2$
26. The line $ax + by + c = 0$ is normal to the circle $x^2 + y^2 + 2gx + 2fy + d = 0$, if :
 (A) $ag + bf + c = 0$ (B) $ag + bf - c = 0$ (C) $ag - bf + c = 0$ (D) $ag - bf - c = 0$
27. A circle passes through $(-2, 4)$ and touches the y -axis at $(0, 2)$. Which one of the following equations can represent a diameter of this circle ?
 (A) $4x + 5y - 6 = 0$ (B) $2x - 3y + 10 = 0$ (C) $3x + 4y - 3 = 0$ (D) $5x + 2y + 4 = 0$
28. Equation of the tangent to the circle, at the point $(1, -1)$, whose centre is the point of intersection of the straight lines $x - y = 1$ and $2x + y = 3$ is :
 (A) $4x + y - 3 = 0$ (B) $x + 4y + 3 = 0$ (C) $3x - y - 4 = 0$ (D) $x - 3y - 4 = 0$
29. If $y + 3x = 0$ is the equation of a chord of the circle, $x^2 + y^2 - 30x = 0$, then the equation of the circle with this chord as diameter is :
 (A) $x^2 + y^2 + 3x + 9y = 0$ (B) $x^2 + y^2 - 3x + 9y = 0$
 (C) $x^2 + y^2 - 3x - 9y = 0$ (D) $x^2 + y^2 + 3x - 9y = 0$
30. If the tangent to the conic, $y - 6 = x^2$ at $(2, 10)$ touches the circle, $x^2 + y^2 + 8x - 2y = k$ (for some fixed k) at a point (α, β) , then (α, β) is :
 (A) $\left(-\frac{6}{17}, \frac{10}{17}\right)$ (B) $\left(-\frac{8}{17}, \frac{2}{17}\right)$ (C) $\left(-\frac{4}{17}, \frac{1}{17}\right)$ (D) $\left(-\frac{7}{17}, \frac{6}{17}\right)$
31. If a circle passing through the point $(-1, 0)$ touches y -axis at $(0, 2)$, then the length of the chord of the circle along the x -axis is :
 (A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) 3 (D) 5
32. If the point $(1, 4)$ lies inside the set circle $x^2 + y^2 - 6x - 10y + p = 0$ and the circle does not touch or intersect the coordinate axes, then the set of all possible values of p is the interval :
 (A) $(0, 25)$ (B) $(25, 39)$ (C) $(9, 25)$ (D) $(25, 29)$

33. The set of all real values of λ for which exactly two common tangents can be drawn to the circles $x^2 + y^2 - 4x - 4y + 6 = 0$ and $x^2 + y^2 - 10x - 10y + \lambda = 0$ is the interval :
 (A) (12, 32) (B) (18, 42) (C) (12, 24) (D) (18, 48)
34. For the two circles $x^2 + y^2 = 16$ and $x^2 + y^2 - 2y = 0$, there is/are :
 (A) three common tangents (B) no common tangent
 (C) two pairs of common tangents (D) one pair of common tangents
35. The equation of the circle described on the chord $3x + y + 5 = 0$ of the circle $x^2 + y^2 = 16$ as diameter is :
 (A) $x^2 + y^2 + 3x + y - 11 = 0$ (B) $x^2 + y^2 + 3x + y + 1 = 0$
 (C) $x^2 + y^2 + 3x + y - 2 = 0$ (D) $x^2 + y^2 + 3x + y - 22 = 0$
36. If each of the lines $5x + 8y = 13$ and $4x - y = 3$ contains a diameter of the circle $x^2 + y^2 - 2(a^2 - 7a + 11)x - 2(a^2 - 6a + 6)y + b^3 + 1 = 0$, then :
 (A) $a = 5$ and $b \notin (-1, 1)$ (B) $a = 2$ and $b \notin (-1, 1)$
 (C) $a = 2$ and $b \in (-\infty, 1)$ (D) $a = 5$ and $b \in (-\infty, 1)$
37. If a circle C passing through $(4, 0)$ touches the circle $x^2 + y^2 + 4x - 6y - 12 = 0$ externally at a point $(1, -1)$, then the radius of the circle C is :
 (A) 5 (B) $2\sqrt{5}$ (C) 4 (D) $\sqrt{57}$
38. If two vertices of an equilateral triangle are $A(-a, 0)$ and $B(a, 0)$, $a > 0$, and the third vertex C lies above x -axis then the equation of the circumcircle of $\triangle ABC$ is :
 (A) $3x^2 + 3y^2 - 2\sqrt{3}ay = 3a^2$ (B) $3x^2 + 3y^2 - 2ay = 3a^2$
 (C) $x^2 + y^2 - 2ay = a^2$ (D) $x^2 + y^2 - \sqrt{3}ay = a^2$
39. If the circle $x^2 + y^2 - 6x - 8y + (25 - a^2) = 0$ touches the axis of x , then a equals :
 (A) 0 (B) ± 4 (C) ± 2 (D) ± 3

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

40. A circle with centre in the first quadrant is tangent to $y = x + 10$, $y = x - 6$ and the Y -axis. Let (p, q) be the centre of the circle. If the value of $(p + q) = a + b\sqrt{a}$, when $a, b \in \mathbb{Q}$. then the value of $|a - b|$ is
41. If the circles $x^2 + y^2 + (3 + \sin \theta)x + 2 \cos \phi y = 0$ and $x^2 + y^2 + (2 \cos \phi)x + 2\lambda y = 0$ touch each other, then the maximum value of λ is
42. Consider the circles C_1 of radius a and C_2 of radius b , $b > a$ both lying in the first quadrant and touching the coordinate axes. C_1 and C_2 touch each other and $\frac{b}{a} = \lambda + \sqrt{\mu}$, $\lambda \in \text{prime number}$ and $\mu \in \text{whole number}$, then $\lambda + \mu$ is

43. The circles $x^2 + y^2 + 2x + 3y + c = 0$ ($c > 0$) and $x^2 + y^2 - x + 2y + c = 0$ intersect orthogonally, then the value of $2c$ is
44. If λ is the radius of a circle having the lines $x^2 + 2xy + 3x + 6y = 0$ as its normals and having size just sufficient to contain the circle $x(x-4) + y(y-3) = 0$, then 2λ is
45. The circle $x^2 + y^2 - 4x - 4y + 4 = 0$ is inscribed in a triangle which has two of its sides along the coordinate axes. The locus of the circumcenter of the triangle is $x + y - xy + k(x^2 + y^2)^{1/2} = 0$. then k is
46. If $(1 + \alpha x)^n = 1 + 8x + 24x^2 + \dots$ and a line through $P(\alpha, n)$ cuts the circle $x^2 + y^2 = 4$ in A and B , then $PA \cdot PB$ is equal to
47. Let $\phi(x, y) = 0$ be the equation of a circle. If $\phi(0, \lambda) = 0$ has equal roots $\lambda = 2, 2$ and $\phi(\lambda, 0) = 0$ has roots $\lambda = \frac{4}{5}, 5$. If the centre of the circle is (α, β) then $10\alpha + 3\beta$ is
48. The point $(1, 4)$ lies inside the circle $x^2 + y^2 - 6x - 10y + \lambda = 0$. If the circle neither touches nor cuts the axes, then the difference between the maximum and the minimum possible values of λ is
49. Consider the family of circles $x^2 + y^2 - 2x - 2\lambda y - 8 = 0$ passing through two fixed points A and B . Then the distance between the points A and B is
50. If a circle $S(x, y) = 0$ touches the point $(2, 3)$ of the line $x + y = 5$ and $S(1, 2) = 0$, then radius of such circle is $\frac{1}{\sqrt{\lambda}}$ units, then the value of λ^2 is .
51. A circle $x^2 + y^2 + 4x - 2\sqrt{2}y + c = 0$ is the director circle of the circle C_1 and C_1 is the director circle of circle C_2 and so on. If the sum of radii of all these circles is 2 and if $c = \lambda\sqrt{2}$, then the value of λ is
52. The length of a common internal tangent of two circles is 5 and that of a common external tangent is 13. If the product of the radii of two circles is λ , then the value of λ is
53. If $ax + by - 5 = 0$ is the equation of the chord of the circle $(x-3)^2 + (y-4)^2 = 4$, which passes through $(2, 3)$ and $(2, 3)$ is at the minimum distance from the centre of the circle, then $a^{2021} + b^{2021}$ is
54. If (a, b) be coordinates of the centre of the smallest circle which cuts the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 - 10x + 12y + 52 = 0$ orthogonally, then $a + b$ is

Complex Numbers

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT.

- If $g(x)$ and $h(x)$ are two polynomials such that the polynomial $P(x) = g(x^3) + xh(x^3)$ is divisible by $x^2 + x + 1$, then which of the following is not correct ?
 (A) $g(1) = h(1) = 0$ (B) $g(1) = h(1) \neq 0$ (C) $g(1) = -h(1)$ (D) $g(1) + h(1) = 0$
- Let $z (\neq 2)$ be a complex number such that $\log_{1/2} |z - 2| > \log_{1/2} |z|$ then :
 (A) $\operatorname{Re}(z) > 1$ (B) $\operatorname{Im}(z) > 1$ (C) $\operatorname{Re}(z) < 1$ (D) $\operatorname{Im}(z) < 1$
- $\frac{\sqrt{5+12i} + \sqrt{5-12i}}{\sqrt{5+12i} - \sqrt{5-12i}} =$
 (A) $-\frac{3}{2}i$ (B) $\frac{3}{2}i$ (C) $-\frac{3}{2}$ (D) $\frac{3}{2}$
- If z_1, z_2 are two complex numbers such that $\operatorname{Im}(z_1 + z_2) = 0, \operatorname{Im}(z_1 z_2) = 0$, then :
 (A) $z_1 = -z_2$ (B) $z_1 = z_2$ (C) $z_1 = \bar{z}_2$ (D) None of these
- The general values of θ which satisfy the equation
 $(\cos \theta + i \sin \theta)(\cos 3\theta + i \sin 3\theta)(\cos 5\theta + i \sin 5\theta) \dots (\cos (2n-1)\theta + i \sin (2n-1)\theta) = 1$ is :
 (where r is an integer)
 (A) $\frac{r\pi}{n^2}$ (B) $\frac{2r\pi}{n^2}$ (C) $\frac{(2r+1)\pi}{n^2}$ (D) None of these
- If $|a_i| < 1, \lambda_i \geq 0$ for $i = 1, 2, 3, \dots, n$ and $\lambda_1 + \lambda_2 + \dots + \lambda_n = 1$ then the value of $|\lambda_1 a_1 + \lambda_2 a_2 + \dots + \lambda_n a_n|$ is :
 (A) Equal to 1 (B) Less than 1 (C) Greater than 1 (D) None of these
- The value of the expression $\left(1 + \frac{1}{\omega}\right)\left(1 + \frac{1}{\omega^2}\right) + \left(2 + \frac{1}{\omega}\right)\left(2 + \frac{1}{\omega^2}\right) + \left(3 + \frac{1}{\omega}\right)\left(3 + \frac{1}{\omega^2}\right) + \dots + \left(n + \frac{1}{\omega}\right)\left(n + \frac{1}{\omega^2}\right)$,
 where ω is a non-zero complex cube root of unity is:
 (A) $\frac{n(n^2+2)}{3}$ (B) $\frac{n(n^2-2)}{3}$ (C) $\frac{n(n^2+1)}{3}$ (D) $\frac{n(n^2-1)}{3}$
- If ω is a non-real cube root of unity, then minimum value of $|a + b\omega + c\omega^2|$ (where a, b, c are integers but not all equal) is :
 (A) 0 (B) $\sqrt{3}/2$ (C) 1 (D) 2
- If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on :
 (A) the real axis (B) the imag. axis (C) a circle (D) a parabola
- If $(a_1 + ib_1)(a_2 + ib_2) \dots (a_n + ib_n) = A + iB$, then $\sum_{k=1}^n \tan^{-1}\left(\frac{b_k}{a_k}\right)$ is equal to :
 (A) $\tan^{-1}\left(\frac{A}{B}\right)$ (B) $\tan^{-1}\left(\frac{B}{A}\right)$ (C) $\left(\frac{A}{B}\right)$ (D) $\left(\frac{B}{A}\right)$

11. Let $z = \cos \theta + i \sin \theta$, then the value of $\sum_{m=1}^{15} \operatorname{Im}(z^{2m-1})$ at $\theta = 2^\circ$ is :
 (A) $\frac{1}{\sin 2^\circ}$ (B) $\frac{1}{2 \sin 2^\circ}$ (C) $\frac{1}{3 \sin 2^\circ}$ (D) $\frac{1}{4 \sin 2^\circ}$
12. If $\cos A + \cos B + \cos C = 0$, $\sin A + \sin B + \sin C = 0$ and $A + B + C = 180^\circ$ then the value of $\cos 3A + \cos 3B + \cos 3C$ is :
 (A) 3 (B) -3 (C) $\sqrt{3}$ (D) $-\sqrt{3}$
13. $A + iB$ form of $\frac{(\cos x + i \sin x)(\cos y + i \sin y)}{(\cot u + i)(1 + i \tan v)}$ is :
 (A) $\sin u \cos v [\cos(x + y - u - v) + i \sin(x + y - u - v)]$
 (B) $\sin u \cos v [\cos(x + y + u + v) + i \sin(x + y + u + v)]$
 (C) $\sin u \cos v [\cos(x + y + u + v) - i \sin(x + y + u + v)]$
 (D) None of these
14. If $\log_{\tan 30^\circ} \left(\frac{2|z|^2 + 2|z| - 3}{|z| + 1} \right) < -2$ then :
 (A) $|z| < \frac{3}{2}$ (B) $|z| > \frac{3}{2}$ (C) $|z| < 2$ (D) $|z| > 2$
15. If $\frac{\tan \alpha - i \left(\sin \frac{\alpha}{2} + \cos \frac{\alpha}{2} \right)}{1 + 2i \sin \frac{\alpha}{2}}$ is purely imaginary, then α is given by :
 (A) $n\pi + \frac{\pi}{4}$ (B) $n\pi - \frac{\pi}{4}$ (C) $(2n + 1)\pi$ (D) $2n\pi + \frac{\pi}{4}$
16. If $\sqrt{1 - c^2} = nc - 1$ and $z = e^{i\theta}$, then $\frac{c}{2n} (1 + nz) \left(1 + \frac{n}{z} \right)$ is equal to :
 (A) $1 - c \cos \theta$ (B) $1 + 2c \cos \theta$ (C) $1 + c \cos \theta$ (D) $1 - 2c \cos \theta$
17. If $z_1 z_2 \in \mathbb{C}$, $z_1^2 + z_2^2 \in \mathbb{R}$, $z_1(z_1^2 - 3z_2^2) = 2$ and $z_2(3z_1^2 - z_2^2) = 11$, then the value of $z_1^2 + z_2^2$ is :
 (A) 5 (B) 6 (C) 10 (D) 12
18. The solution of the equation $|z| - z = 1 + 2i$ is
 (A) $2 - \frac{3}{2}i$ (B) $\frac{3}{2} + 2i$ (C) $\frac{3}{2} - 2i$ (D) $-2 + \frac{3}{2}i$

PARAGRAPH FOR QUESTIONS 19 - 21

Consider the complex numbers z_1 and z_2 satisfying the relation $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$

19. Complex number $z_1 \bar{z}_2$ is :
 (A) purely real (B) purely imaginary (C) zero (D) None of these
20. Complex number z_1 / z_2 is :
 (A) purely real (B) purely imaginary (C) zero (D) None of these

21. One of the possible argument of complex number $i\left(\frac{z_1}{z_2}\right)$:
- (A) $\pi/2$ (B) $-\pi/2$ (C) 0 (D) None of these
22. The largest value of r for which the region represented by the set $\{\omega \in C / |\omega - 4 - i| \leq r\}$ is contained in the region represented by the set $\{\omega \in C / |z - 1| \leq |z + i|\}$, is equal to :
- (A) $\sqrt{17}$ (B) $2\sqrt{2}$ (C) $\frac{3}{2}\sqrt{2}$ (D) $\frac{5}{2}\sqrt{2}$
23. If $Z_1 \neq 0$ and Z_2 be two complex numbers such that $\frac{Z_2}{Z_1}$ is a purely imaginary number, then $\left|\frac{2Z_1 + 3Z_2}{2Z_1 - 3Z_2}\right|$ is equal to:
- (A) 2 (B) 5 (C) 3 (D) 1
24. If a complex number z satisfies the equation $z + \sqrt{2}|z + 1| + i = 0$, then $|z|$ is equal to :
- (A) 2 (B) $\sqrt{3}$ (C) $\sqrt{5}$ (D) 1
25. The point represented by $2 + i$ in the Argand plane moves 1 unit eastwards, then 2 units northwards and finally from there $2\sqrt{2}$ units in the south-westwards direction. Then its new position in the Argand plane is at the point represented by :
- (A) $2 + 2i$ (B) $1 + i$ (C) $-1 - i$ (D) $-2 - 2i$
26. Let $z = 1 + ai$ be a complex number, $a > 0$, such that z^3 is a real number. Then the sum $1 + z + z^2 + \dots + z^{11}$ is equal to :
- (A) $-1250\sqrt{3}i$ (B) $1250\sqrt{3}i$ (C) $1365\sqrt{3}i$ (D) $-1365\sqrt{3}i$
27. Let w ($\text{Im } w \neq 0$) be a complex number. Then the set of all complex numbers z satisfying the equation $w - \bar{w}z = k(1 - z)$, for some real number k , is :
- (A) $\{z : |z| = 1\}$ (B) $\{z : z = \bar{z}\}$ (C) $\{z : z \neq \bar{z}\}$ (D) $\{z : |z| = 1, z \neq 1\}$
28. If z_1, z_2 and z_3, z_4 are 2 pairs of complex conjugate numbers, then $\arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right)$ equals :
- (A) 0 (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{2}$ (D) π
29. Let $z \neq -i$ be any complex number such that $\frac{z-i}{z+i}$ is a purely imaginary number. Then $z + \frac{1}{z}$ is :
- (A) 0 (B) any non-zero real number
(C) a purely imaginary number (D) any non-zero real number other than 1
30. For all complex numbers z of the form $1 + i\alpha$, $\alpha \in R$, if $z^2 = x + iy$, then :
- (A) $y^2 - 4x + 2 = 0$ (B) $y^2 + 4x - 4 = 0$ (C) $y^2 - 4x + 4 = 0$ (D) $y^2 + 4x + 2 = 0$
31. Let $a = \text{Im}\left(\frac{1+z^2}{2iz}\right)$, where z is any non-zero complex number. The set $A = \{a : |z| = 1 \text{ and } z \neq \pm 1\}$ is equal to :
- (A) $(-1, 1)$ (B) $[-1, 1]$ (C) $[0, 1)$ (D) $(-1, 0]$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

32. If z is a complex number satisfying the equation $|z+i| + |z-i| = 8$ on the complex plane. Let m and M be the smallest and the largest value of $|z|$ respectively, then $[m+M] =$ ($[.]$ denotes greatest integer function)
33. Let α be a complex number such that $|\alpha| = 1$. If the equation $\alpha z^2 + z + 1 = 0$ has a purely imaginary root, then $\tan^2(\arg \alpha) = \frac{a+\sqrt{b}}{c}$, where a, b, c are coprime natural number, then $a+b+c =$
34. Let $a = \sum_{r=1}^8 \tan^2\left(\frac{r\pi}{17}\right)$ and $b = \prod_{r=1}^8 \tan^2\left(\frac{r\pi}{17}\right)$, then $\frac{a}{b}$ is equal to:
35. Consider the set, $A = \{z \in \mathbb{C} \mid z = x-1+xi, x \in \mathbb{R}\}$ Find the number of complex numbers $z, z \in A$ such that $|z| \leq |\omega|$ for all $\omega \in A$.
36. Let $z \in \mathbb{C}$ with $\operatorname{Re}(z) > 1$, then $\left|\frac{1}{z} - \frac{1}{2}\right| < L$. Find the smallest value of L .
37. If $z + \frac{1}{z} = \sqrt{3}$, then $\sum_{r=1}^5 \left(z^r + \frac{1}{z^r}\right)^2 =$
38. If z is the imaginary 5th root of unity, then $\frac{z^4}{1+z^4} + \frac{z}{1+z} + \frac{z^2}{1+z^2} + \frac{z^4}{z+z^4} =$
39. The sets $A = \{z \mid z^{18} = 1\}$ and $B = \{\omega \mid \omega^{48} = 1\}$ consider the set $C = \{Z\omega \mid z \in A \text{ and } \omega \in B\}$. How many distinct elements are in C ?
40. A function f is defined on the complex numbers by $f(z) = (a+bi)z$, where a and b are real number, This function has the property that image of each point in the complex plane is equidistant from that point and the origin. Given that $|a+bi| = 8$ and that $b^2 = \frac{m}{n}$, where m and n are relatively prime positive integers, find $m+n$.
41. There is a complex number z with imaginary part 164 and a positive integer n such that $\frac{z}{z+n} = 4i$. Find n .
42. The polynomial $f(z) = az^{2018} + bz^{2017} + cz^{2016}$ has real coefficients not exceeding 2019 and $f\left(\frac{1+\sqrt{3}i}{2}\right) = 2015 + 2019\sqrt{3}i$. Find the remainder when $f(1)$ is divided by 1000.
43. If the number of solutions of the equation $z^{2017} = p\bar{z}$, where p is a non zero given complex number is equal to n then find the value of $n - 2000$.
44. Let $A(z_1), B(z_2), C(z_3), D(z_4)$ be vertices of a square, satisfy $|z_1-1| = |z_2-1| = |z_3-1| = |z_4-1|$. Then $z_1 + z_2 + z_3 + z_4 =$
45. If the imaginary part of $(1-i)^n(1+i)^n$ be negative (where $n \in \mathbb{N}, n < 100$), then if the sum of all the possible values of n is α find the value of $\alpha/5$.
46. If $|z-i| \leq 2$ and $z_0 = 5+3i$, then the sum of maximum and minimum values of $|iz+z_0| =$

Permutations & Combinations

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

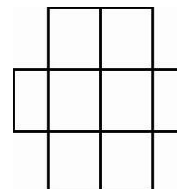
- In how many ways 6 letters can be placed in 6 envelopes such that no letter is placed in its corresponding envelope.
(A) 265 (B) 275 (C) 255 (D) None of these
- Find the sum of all four digits numbers that can be formed using the digits 0, 1, 2, 3, 4 ; no digits being repeated in any number.
(A) 259980 (B) 249980 (C) 269960 (D) None of these
- In a plane there are 37 straight line, of which 13 pass through the point A and 11 pass through the point B. Besides, no three lines pass through one point, no line passes through both points A and B, and no two are parallel. Find the number of points of intersection of the straight lines.
(A) 533 (B) 536 (C) 535 (D) 530
- There are p points in a plane, no three of which are in the same straight line with the exception of q, which are all in the same straight line. Straight lines which can be formed by joining them :
(A) ${}^pC_2 - {}^qC_2$ (B) ${}^pC_2 - {}^qC_2 + 1$ (C) ${}^pC_2 + {}^qC_2$ (D) ${}^pC_2 + {}^qC_2 + 1$
- 10 different toys are to be distributed among 10 children. Total number of ways of distributing these toys so that exactly 2 children do not get any toy, is equal to :
(A) $(10!)^2 \left(\frac{1}{3!2!7!} + \frac{1}{(2!)^5 6!} \right)$ (B) $(10!)^2 \left(\frac{1}{3!2!7!} + \frac{1}{(2!)^4 6!} \right)$
(C) $(10!)^2 \left(\frac{1}{3!7!} + \frac{1}{(2!)^5 6!} \right)$ (D) $(10!)^2 \left(\frac{1}{3!7!} + \frac{1}{(2!)^4 6!} \right)$
- In how many ways we can divide 52 playing cards among 4 players equally ?
(A) $\frac{52!}{(13!)^4} \times \frac{1}{4}$ (B) $\frac{52!}{(13!)^4}$ (C) $\frac{52!4}{(13!)^4}$ (D) None of these
- In how many ways can a party of 4 men and 4 women be seated at a circular table so that no two women are adjacent ?
(A) 144 (B) 24 (C) 72 (D) 288

PARAGRAPH FOR QUESTIONS 8 - 9

A round table conference is to be held between 20 delegates of 2 countries.

- In how many ways can they be seated if two particular delegates are always together ?
(A) 18! (B) $2 \times 18!$ (C) 17! (D) $2 \times 17!$
- In how many ways can they be seated if two particular delegates are never together ?
(A) $17 \times 18!$ (B) $18 \times 18!$ (C) $15 \times 18!$ (D) $16 \times 18!$
- If the number of ways of selecting k coupons one by one out of an unlimited number of coupons bearing the letters A, T, M so that they cannot be used to spell the word MAT is 93, then k equals to :
(A) 3 (B) 5 (C) 7 (D) 8

11. Total number of positive integral solution of $15 < x_1 + x_2 + x_3 \leq 20$, is equal to :
 (A) 1125 (B) 1150 (C) 1245 (D) 685
12. The number of permutations of the letters of the HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears, are :
 (A) 166674 (B) 169194 (C) 166680 (D) 181434
13. If $3n$ different things can be equally distributed among 3 persons in k ways then the number of ways to divide the $3n$ things in 3 equal groups is :
 (A) $k \times 3!$ (B) $\frac{k}{3!}$ (C) $(3!) k$ (D) $3k$
14. Let A be the set of 4-digit numbers $a_1 a_2 a_3 a_4$ where $a_1 > a_2 > a_3 > a_4$ then $n(A)$ is equal to :
 (A) 126 (B) 84 (C) 210 (D) 120
15. A shopkeeper sells three varieties of perfumes and he has a large number of bottles of the same size of each variety in his stock. There are 5 places in row in his showcase. The number of different ways of displaying the three varieties of perfumes in the showcase is :
 (A) 6 (B) 50 (C) 150 (D) 90
16. The number of subsets of the set $A = \{a_1, a_2, \dots, a_n\}$ which contain even number of elements is :
 (A) 2^{n-1} (B) $2^n - 1$ (C) 2^{n-2} (D) 2^n
17. How many numbers are there between 100 and 1000 in which all the digits are distinct ?
 (A) 648 (B) 729 (C) 576 (D) 810
18. Six X have to be placed in the squares of figure such that each row contains at least one X. The number of ways in which this can be done is :
 (A) 25 (B) 26 (C) 27 (D) 30
19. If all the letters of the word 'AGAIN' be arranged as in a dictionary, what is the fiftieth word?
 (A) NAAIG (B) NAAGI (C) NAGAI (D) NAGIA
20. The range of the function ${}^{7-x}P_{x-3}$ is :
 (A) $\{1, 2, 3, 4\}$ (B) $\{1, 2, 3, 4, 5, 6\}$ (C) $\{1, 2, 3\}$ (D) $\{1, 2, 3, 4, 5\}$
21. The results of 21 football matches (win, lose or draw) are to be predicted. The number of forecasts that contain exactly 18 correct results is :
 (A) ${}^{21}C_3 2^{18}$ (B) ${}^{21}C_{18} 2^3$ (C) $3^{21} - 2^{18}$ (D) ${}^{21}C_3 3^{21} - 2^{18}$
22. Letters of the word INDIALOIL are arranged in all possible ways. The number of permutations in which A, I, O occur only at odd places, is :
 (A) 720 (B) 360 (C) 240 (D) 120
23. At an election there are five candidates and three members are to be elected, and a voter may vote for any number of candidates not greater than the number to be elected. The number of ways in which the person can vote is :
 (A) 25 (B) 30 (C) 35 (D) $2^5 - 2^3$



24. A five digit number divisible by 3 is to be formed using the digits 0, 1, 2, 3, 4 and 5 without repetition. The total number of ways, in which this can be done is :
 (A) 240 (B) 3125 (C) 600 (D) 216
25. The number of ways in which two teams A and B of 11 players each can be made up from 22 players so that two particular players are on the opposite sides is :
 (A) 369512 (B) 184755 (C) 184756 (D) 369514
26. A library has a copies of one book, b copies of each of two books, c copies of each of three books and single copies of d books. The total number of ways in which these books can be distributed is :
 (A) $\frac{(a+b+c+d)!}{a! b! c!}$ (B) $\frac{(a+2b+3c+d)!}{a! (b!)^2 (c!)^3}$
 (C) $\frac{(a+2b+3c+d)!}{a! b! c!}$ (D) None of these
27. Given that n is odd, the number of ways in which three numbers in AP can be selected from $1, 2, 3, \dots, n$ is :
 (A) $\frac{(n-1)^2}{2}$ (B) $\frac{(n+1)^2}{4}$ (C) $\frac{(n+1)^2}{2}$ (D) $\frac{(n-1)^2}{4}$
28. 20 persons are invited for a party. In how many different ways can they and the host be seated at circular table, if the two particular persons are to be seated on either side of the host ?
 (A) $\underline{20}$ (B) $\underline{2 \times 18}$ (C) $\underline{18}$ (D) None of these
29. Assuming that no two consecutive digits are same, the number of n -digit numbers is :
 (A) $\underline{n}$ (B) $\underline{9}$ (C) 9^n (D) n^9
30. In how many ways can 21 identical English and 19 identical Hindi books be placed in a row so that no two Hindi books are together ?
 (A) 1540 (B) 1450 (C) 1504 (D) 1405
31. If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 3 : 5$, then n equals :
 (A) 5 (B) 4 (C) 3 (D) 2
32. The greatest number of points of intersection of n circles and m straight lines is :
 (A) $2mn + {}^mC_2$ (B) $\frac{1}{2}m(m-1) + n(2m+n-1)$
 (C) ${}^mC_2 + 2({}^nC_2)$ (D) None of these
33. There are three pigeon holes marked M, P, C . The number of ways in which we can put 12 letters so that 6 of them are in M , 4 are in P and 2 are in C is :
 (A) 2520 (B) 13860 (C) 12530 (D) 25220
34. Three straight lines l_1, l_2 and l_3 are parallel and lie in the same plane. Five points are taken on each of l_1, l_2 and l_3 . The maximum number of triangles which can be obtained with vertices at these points, is :
 (A) 425 (B) 405 (C) 415 (D) 505
35. There are 15 points in a plane, no three of which are in a straight line except 4, all of which are in a straight line. The number of triangles that can be formed by using these 15 points is :
 (A) 404 (B) 415 (C) 451 (D) 490

36. If the four letter words (need not be meaningful) are to be formed using the letters from the word "MEDITERRANEAN" such that the first letter is R and the fourth letter is E , then the total number of all such words is :
- (A) $\frac{11!}{(2!)^3}$ (B) 110 (C) 56 (D) 59
37. If $\frac{{}^{n+2}C_6}{{}^{n-2}P_2} = 11$, then n satisfies the equation :
- (A) $n^2 - 3n - 108 = 0$ (B) $n^2 + 5n - 84 = 0$
 (C) $n^2 + 2n - 80 = 0$ (D) $n^2 + n - 110 = 0$
38. The number of ways of selecting 15 teams from 15 men and 15 women, such that each team consists of a man and a woman, is :
- (A) 1120 (B) 1240 (C) 1880 (D) 1960
39. Let $A = \{x_1, x_2, \dots, x_7\}$ and $B = \{y_1, y_2, y_3\}$ be two sets containing seven and three distinct elements respectively. Then the total number of functions $f : A \rightarrow B$ that are onto, if there exists exactly three elements x in A such that $f(x) = y_2$, is equal to :
- (A) $14 \cdot {}^7C_2$ (B) $16 \cdot {}^7C_3$ (C) $12 \cdot {}^7C_2$ (D) $14 \cdot {}^7C_3$
40. The sum of the digits in the unit's place of all the 4-digits numbers formed by using the numbers 3, 4, 5 and 6, without repetition is :
- (A) 432 (B) 108 (C) 36 (D) 18
41. An eight digit number divisible by 9 is to be formed using digits from 0 to 9 without repeating the digits. The number of ways in which this can be done is :
- (A) $72(7!)$ (B) $18(7!)$ (C) $40(7!)$ (D) $36(7!)$
42. Two women and some men participated in a chess tournament in which every participant played two games with each of the other participants. If the number of games that the men played between themselves exceeds the number of games that the men played with the women by 66, then the number of men who participated in the tournament lies in the interval :
- (A) $[8, 9]$ (B) $[10, 12]$ (C) $(11, 13]$ (D) $(14, 17)$
43. The number of ways in which an examiner can assign 30 marks to 8 questions, giving not less than 2 marks to any question, is :
- (A) ${}^{30}C_7$ (B) ${}^{21}C_7$ (C) ${}^{21}C_7$ (D) ${}^{30}C_8$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

44. All possible three digits even numbers which can be formed with the condition that if 5 is one of the digit then 7 is the next digit is:
45. The number of arrangements of the letters 'a b c d' in which neither a, b nor c, d come together is:
46. The number of three-digit numbers having only two consecutive digits identical is
47. A committee of 5 is to be chosen from a group of 9 people. Number of ways in which it can be formed if two particular persons either serve together or not at all and two other particular persons refuse to serve with each other, is
48. If m denotes the number of 5-digit numbers if each successive digits are in their descending order of magnitude and n is the corresponding figure, when the digits are in their ascending order of magnitude then $(m - n)$ has the value
49. A rack has 5 different pairs of shoes. The number of ways in which 4 shoes can be chosen from it so that there will be no complete pair is
50. An old man while dialing a 7-digit telephone number remembers that the first four digits consists of one 1's, one 2's and two 3's. He also remembers that the fifth digit is either a 4 or 5 while has no memorising of the sixth digit, he remembers that the seventh digit is 9 minus the sixth digit. Maximum number of distinct trials he has to try to make sure that he dials the correct telephone number, is:
51. Number of three-digit number with atleast one 3 and at least one 2 is
52. In a unique hockey series between India & Pakistan, they decide to play on till a team wins 5 matches. The number of ways in which the series can be won by India, if no match ends in a draw is:
53. Define a 'good word' as a sequence of letters that consists only of the letters A, B and C and in which A never immediately followed by B, B is never immediately followed by C, and C is never immediately followed by A. If the number of n-letter good words are 384, find the value of n.
54. Fifty college teachers are surveyed as to their possession of colour TV, VCR and tape recorder. Of them, 22 own colour TV, 15 own VCR and 14 own tape recorders. Nine of these college teachers own exactly two items out of colour TV, VCR and tape recorders; and, one college teacher owns all three, how many of the 50 college teachers own none of three, colour TV, VCR or tape recorder?
55. Six people are going to sit in a row on a bench. A and B are adjacent, C does not want to sit adjacent to D. E and F can sit anywhere. Number of ways in which these six people can be seated, is
56. Find the number of 5-digit numbers using the digits 0, 1, 2, 3, 4 without repetition which are divisible by 4.
57. One hundred management students who read at least one of the three business magazines are surveyed to study the readership pattern. It is found that 80 read Business India, 50 read Business world, and 30 read Business Today. Five students read all the three magazines. How many read exactly two magazines?
58. Number of positive integer solutions satisfying the equation $(x_1 + x_2 + x_3)(y_1 + y_2) = 77$, is _____.

Binomial Theorem

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- The coefficient of x^{20} in the expansion of $(1+x^2)^{40} \cdot \left(x^2 + 2 + \frac{1}{x^2}\right)^{-5}$ is :
 (A) ${}^{30}C_{10}$ (B) ${}^{30}C_{25}$ (C) 1 (D) None of these
- The value of $\sum_{r=0}^n (-1)^r {}^nC_r \frac{1+r \log_e 10}{(1+\log_e 10^n)^r}$ equals :
 (A) 1 (B) -1 (C) n (D) 0
- The coefficient of x^5 in the expression of $(1+x)^{21} + (1+x)^{22} + \dots + (1+x)^{30}$ is :
 (A) ${}^{51}C_5$ (B) 9C_5 (C) ${}^{31}C_6 - {}^{21}C_6$ (D) ${}^{30}C_5 + {}^{20}C_5$
- If $R = (\sqrt{2} + 1)^{2n+1}$ and $f = R - [R]$, where $[]$ denotes the greatest integer function, then $[R]$ equals :
 (A) $f + \frac{1}{f}$ (B) $f - \frac{1}{f}$ (C) $\frac{1}{f} - f$ (D) None of these
- If $R = (7 + 4\sqrt{3})^{2n} = I + f$, where $I \in N$, and $0 \leq f < 1$, then $R(1-f)$ equals :
 (A) $(7 - 4\sqrt{3})^{2n}$ (B) $\frac{1}{(7 + 4\sqrt{3})^{2n}}$ (C) 1 (D) None of these
- Let $(1+x)^n = \sum_{r=0}^n a_r x^r$. Then $\left(1 + \frac{a_1}{a_0}\right)\left(1 + \frac{a_2}{a_1}\right) \dots \left(1 + \frac{a_n}{a_{n-1}}\right)$ is equal to :
 (A) $\frac{(n+1)^{n+1}}{n!}$ (B) $\frac{(n+1)^n}{n!}$ (C) $\frac{n^{n-1}}{(n-1)!}$ (D) $\frac{(n+1)^{n+1}}{(n-1)!}$
- If $n > 3$, then
 $xyC_0 - (x-1)(y-1)C_1 + (x-2)(y-2)C_2 - (x-3)(y-3)C_3 + \dots + (-1)^n (x-n)(y-n)C_n$ equals:
 (A) $xy \times 2^n$ (B) nxy (C) xy (D) None of these
- If n is an odd natural number, then $\sum_{r=0}^n \frac{(-1)^r}{{}^nC_r}$ equals :
 (A) 0 (B) $\frac{1}{n}$ (C) $\frac{n}{2^n}$ (D) None of these
- The value of $\sum_{r=0}^n (-2)^r \left(\frac{{}^nC_r}{r+2} C_r\right)$ is :
 (A) $\frac{1}{n+1}$, n is odd (B) $\frac{1}{n+2}$, n is even (C) $\frac{1}{n+1}$, n is even (D) None of these
- In the binomial expansion of $(a-b)^n$, $n \geq 5$, the sum of the 5th and 6th terms is zero. The a/b equals :
 (A) $\frac{n-5}{6}$ (B) $\frac{n-4}{5}$ (C) $\frac{5}{n-4}$ (D) $\frac{6}{n-5}$

11. The coefficient of x^{53} in $\sum_{r=0}^{100} {}^{100}C_r (x-3)^{100-r} 2^r$ is :
 (A) ${}^{100}C_{51}$ (B) ${}^{100}C_{52}$ (C) $-{}^{100}C_{53}$ (D) ${}^{100}C_{54}$
12. The value of $\frac{{}^{47}C_4}{{}^{57}C_4} + \sum_{j=0}^3 \frac{{}^{50-j}C_3}{{}^{57}C_{53}} + \sum_{j=0}^5 \frac{{}^{56-k}C_{53-k}}{{}^{57}C_4}$ is :
 (A) 0 (B) 2 (C) 1 (D) None of these
13. The value of $\frac{1}{81^n} - \frac{10}{81^n} {}^{2n}C_1 + \frac{10^2}{81^n} {}^{2n}C_2 - \frac{10^3}{81^n} {}^{2n}C_3 + \dots + \frac{10^{2n}}{81^n}$ is :
 (A) 2 (B) 0 (C) 1/2 (D) 1
14. If the magnitude of the coefficient of x^7 in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^8$, where a, b are positive numbers, is equal to the magnitude of the coefficient of x^{-7} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^8$, then a and b are connected by the relation :
 (A) $ab = 1$ (B) $ab = 2$ (C) $a^2b = 1$ (D) $ab^2 = 2$
15. The coefficient of t^{24} in the expansion of $(1+t^2)^{12} (1+t^{12}) (1+t^{24})$ is :
 (A) $12C_6 + 2$ (B) $12C_5$ (C) $12C_6$ (D) $12C_7$
16. The remainder obtained when $(1!)^2 + (2!)^2 + (3!)^2 + \dots + (100!)^2$ is divided by 10^2 is :
 (A) 27 (B) 28 (C) 17 (D) 14
17. The value of $({}^7C_0 + {}^7C_1) + ({}^7C_1 + {}^7C_2) + \dots + ({}^7C_6 + {}^7C_7)$ is :
 (A) $2^8 - 1$ (B) $2^8 + 1$ (C) 2^8 (D) $2^8 - 2$
18. If $(1+x)^{15} = a_0 + a_1x + \dots + a_{15}x^{15}$, then $\sum_{r=1}^{15} r \frac{a_r}{a_{r-1}}$ is equal to :
 (A) 110 (B) 115 (C) 120 (D) 135
19. ${}^{n-2}C_r + 2 {}^{n-2}C_{r-1} + {}^{n-2}C_{r-2}$ equals :
 (A) ${}^{n+1}C_r$ (B) nC_r (C) ${}^nC_{r+1}$ (D) ${}^{n-1}C_r$
20. The value of ${}^{15}C_8 + {}^{15}C_9 - {}^{15}C_6 - {}^{15}C_7$ is :
 (A) -1 (B) 0 (C) 1 (D) None of these
21. If ${}^{18}C_{15} + 2({}^{18}C_{16}) + {}^{17}C_{16} + 1 = {}^nC_3$, then n is equal to
 (A) 19 (B) 20 (C) 18 (D) 24
22. For $x \in \mathbb{R}, x \neq -1$, if $(1+x)^{2016} + x(1+x)^{2015} + x^2(1+x)^{2014} + \dots + x^{2016} = \sum_{i=0}^{2016} a_i x^i$, then a_{17} is equal to :
 (A) $\frac{2017!}{17! 2000!}$ (B) $\frac{2016!}{17! 1999!}$ (C) $\frac{2017!}{2000!}$ (D) $\frac{2016!}{16!}$

23. The value of $\sum_{r=1}^{15} r^2 \left(\frac{{}^{15}C_r}{{}^{15}C_{r-1}} \right)$ is equal to :
 (A) 560 (B) 680 (C) 1240 (D) 1085
24. If the coefficients of the three successive terms in the binomial expansion of $(1+x)^n$ are in the ratio 1 : 7 : 42, then the first of these terms in the expansion is :
 (A) 6th (B) 7th (C) 8th (D) 9th
25. The number of terms in the expansion of $(1+x)^{101} (1+x^2-x)^{100}$ in powers of x is :
 (A) 302 (B) 301 (C) 202 (D) 101
26. If $\left(2 + \frac{x}{3}\right)^{55}$ is expanded in the ascending powers of x and the coefficients of powers of x in two consecutive terms of the expansion are equal, then these terms are :
 (A) 7th and 8th (B) 8th and 9th (C) 27th and 28th (D) 28th and 29th
27. The ratio of the coefficient of x^{15} to the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^{15}$ is :
 (A) 7 : 16 (B) 7 : 64 (C) 1 : 4 (D) 1 : 32
28. If the 7th term in the binomial expansion of $\left(\frac{3}{\sqrt[3]{84}} + \sqrt{3} \ln x\right)^9$, $x > 0$, is equal to 729, then x can be :
 (A) e^2 (B) e (C) $\frac{e}{2}$ (D) $2e$
29. The sum of the rational terms in the binomial expansion of $\left(2^{\frac{1}{2}} + 3^{\frac{1}{5}}\right)^{10}$ is :
 (A) 25 (B) 32 (C) 9 (D) 41
30. If the coefficient of x^{-2} and x^{-4} in the expansion of $\left(x^{\frac{1}{3}} + \frac{1}{2x^{\frac{1}{3}}}\right)^{18}$, $(x > 0)$, are m and n respectively, then $\frac{m}{n}$ is equal to :
 (A) 182 (B) $\frac{4}{5}$ (C) $\frac{5}{4}$ (D) 27
31. The term independent of x in the binomial expansion of $\left(1 - \frac{1}{x} + 3x^5\right) \left(2x^2 - \frac{1}{x}\right)^8$ is :
 (A) 400 (B) 496 (C) -400 (D) -496
32. The coefficient of x^{50} in the binomial expansion of $(1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \dots + x^{1000}$ is :
 (A) $\frac{(1000)!}{(50)!(950)!}$ (B) $\frac{(1000)!}{(49)!(951)!}$ (C) $\frac{(1001)!}{(51)!(950)!}$ (D) $\frac{(1001)!}{(50)!(951)!}$
33. If $1+x^4+x^5 = \sum_{i=0}^5 a_i (1+x)^i$, for all x in R , then a_2 is :
 (A) 6 (B) 10 (C) -4 (D) -8

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

34. If the coefficients of the $(2r+4)^{\text{th}}, (r-2)^{\text{th}}$ terms in the expansion of $(1+x)^{18}$ are equal, find r .
35. Find the coefficient of x^5 in the expansion of $(1+x^2)^5 \cdot (1+x)^4$ is.
36. Find the coefficient of x^4 in the expansion of $(1+x+x^2+x^3)^{11}$
37. Let $(1+x^2)^2 \cdot (1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$. If a_1, a_2 and a_3 are in A.P. find sum of possible values of n .
38. If $(7+4\sqrt{3})^n = p + \beta$ when n & p are positive integers and β is a proper fraction. Find the value of $(1-\beta)(p+\beta)$.
39. Let $a = 3^{1/223} + 1$ and for all $n \geq 3$, let $f(n) = {}^nC_0 \cdot a^{n-1} - {}^nC_1 \cdot a^{n-2} + {}^nC_2 \cdot a^{n-3} - \dots + (-1)^{n-1} \cdot {}^nC_{n-1} \cdot a^0$. If the value of $f(2007) + f(2008) = 81k$ where $k \in N$, then find k .
40. If the value of $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} - \dots + \binom{30}{20}\binom{30}{30}$ is x_{C_y} where $y \leq 10$ then find $x + y$.
41. Find the coefficient of x^3 in $\left(ax + \frac{b}{x^2}\right)^9$ and x^{-3} in $\left(ax^2 - \frac{b}{x}\right)^9$. Find the value of $a + b$ so that their coefficient are equal:
42. In the binomial expansion of $(1+x)^n$, the coefficient of the fifth, sixth and seventh terms are in A.P. Find product of all possible values of n .
43. How many terms are free from radical sign in the expansion of $(x^{1/2} + y^{1/3})^{100}$.
44. If $(1+ax)^n = 1+8x+24x^2+\dots$, then find $a + n$.
45. Find the number of integral terms in the expansion of $(\sqrt{3} + \sqrt[8]{5})^{256}$ is
46. If $|x| < 1$, then find the coefficient of x^2 in the expansion of $(1+2x+3x^2+4x^3+\dots)^2$.
47. Find the value of $7k$ for which the coefficients of the middle terms in the expansions of $(1+kx)^6$ and $(1-kx)^8$ are equal.
48. Find the term independent of x in the expansion of $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$.

Trigonometric Identities & Equations

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- The equation $a \sin x + \cos 2x = 2a - 7$ possesses a solution if :
 (A) $a > 6$ (B) $2 \leq a \leq 6$ (C) $a > 2$ (D) None of these
- If $\tan^2 x + \sec x - a = 0$ has at least one solution, then complete set of values of a is :
 (A) $(-\infty, 1]$ (B) $[-1, \infty)$ (C) $\left[\frac{9}{4}, \infty\right)$ (D) $[1, \infty)$
- If $\sin(\theta - x) = a$, $\cos(\theta - y) = b$, then $\cos(x - y) =$
 (A) $a\sqrt{1-b^2} + b\sqrt{1-a^2}$ (B) ab
 (C) $a\sqrt{1-b^2} - b\sqrt{1-a^2}$ (D) $2ab$
- The product $\cot 123^\circ \cdot \cot 133^\circ \cdot \cot 137^\circ \cdot \cot 147^\circ$, when simplified is equal to :
 (A) -1 (B) $\tan 37^\circ$ (C) $\cot 33^\circ$ (D) 1
- If $(1 + \tan 1^\circ) \cdot (1 + \tan 2^\circ) \cdot (1 + \tan 3^\circ) \dots (1 + \tan 45^\circ) = 2^n$, then ‘ n ’ is equal to :
 (A) 16 (B) 23 (C) 30 (D) None of these
- If $4^{\sin 2x + 2\cos^2 x} + 4^{1 - \sin 2x + 2\sin^2 x} = 65$, then $(\sin 2x + \cos 2x)$ has the value equal to :
 (A) -1 (B) 2 (C) $\sqrt{2}$ (D) 1
- If $P = \cos \frac{\pi}{20} \cdot \cos \frac{3\pi}{20} \cdot \cos \frac{7\pi}{20} \cdot \cos \frac{9\pi}{20}$ and $Q = \cos \frac{\pi}{11} \cdot \cos \frac{2\pi}{11} \cdot \cos \frac{4\pi}{11} \cdot \cos \frac{8\pi}{11} \cdot \cos \frac{16\pi}{11}$, then $\frac{P}{Q}$ is :
 (A) Not defined (B) 1 (C) 2 (D) None of these
- The solution of the equation $4\sin^4 x + \cos^4 x = 1$ is :
 (A) $x = 2n\pi$ (B) $x = n\pi + \frac{\pi}{2}$ (C) $x = n\pi$ (D) None of these
- If A, B, C, D are the angles of a cyclic quadrilateral, then $\cos A + \cos B + \cos C + \cos D =$
 (A) 4 (B) 1 (C) 0 (D) -1
- In a ΔPQR , if $3\sin P + 4\cos Q = 6$ and $4\sin Q + 3\cos P = 1$, then the angle R is equal to :
 (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{6}$ (D) $\frac{\pi}{6}$
- If $\tan \theta_1 = k \cot \theta_2$ then $\frac{\cos(\theta_1 - \theta_2)}{\cos(\theta_1 + \theta_2)} =$
 (A) $\frac{1+k}{1-k}$ (B) $\frac{1-k}{1+k}$ (C) $\frac{k+1}{k-1}$ (D) $\frac{k-1}{k+1}$
- $\frac{1}{\cos \alpha + \cos 3\alpha} + \frac{1}{\cos \alpha + \cos 5\alpha} + \dots + \frac{1}{\cos \alpha + \cos(2n+1)\alpha} =$
 (A) $\operatorname{cosec} \alpha [\tan(n+1)\alpha - \tan \alpha]$ (B) $\sec \alpha [\tan(n+1)\alpha - \tan \alpha]$
 (C) $\frac{1}{2} \sec \alpha [\tan(n+1)\alpha - \tan \alpha]$ (D) $\frac{1}{2} \operatorname{cosec} \alpha [\tan(n+1)\alpha - \tan \alpha]$

13. The range of $\log_{\sqrt{5}} [\sqrt{2}(\sin x - \cos x) + 3]$ is :
 (A) $[0, 2]$ (B) $[1, 2]$ (C) $[0, 3]$ (D) $[1, 3]$
14. $\cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{14\pi}{15} =$
 (A) $1/4$ (B) $1/8$ (C) $1/16$ (D) $1/32$
15. If $|\tan x| \leq 1$ and $x \in [-\pi, \pi]$ then the exhaustive solution set for x is :
 (A) $\left[-\pi, -\frac{3\pi}{4}\right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \pi\right]$ (B) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \pi\right]$
 (C) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ (D) None of these
16. If $\cos \theta = \frac{1}{2} \left(x + \frac{1}{x}\right)$, then $\frac{1}{2} \left(x^2 + \frac{1}{x^2}\right)$ is equal to:
 (A) $\sin 2\theta$ (B) $\cos 2\theta$ (C) $\tan 2\theta$ (D) $\sec 2\theta$
17. If $\tan x = \frac{b}{a}$, then $\sqrt{\frac{a+b}{a-b}} + \sqrt{\frac{a-b}{a+b}}$ is equal to :
 (A) $\frac{2 \sin x}{\sqrt{\sin 2x}}$ (B) $\frac{2 \cos x}{\sqrt{\cos 2x}}$ (C) $\frac{2 \cos x}{\sqrt{\sin 2x}}$ (D) $\frac{2 \sin x}{\sqrt{\cos 2x}}$
- *18. If $\sec \theta + \tan \theta = 1$, then root of the equation $(a - 2b + c)x^2 + (b - 2c + a)x + (c - 2a + b) = 0$ is :
 (A) $\sec \theta$ (B) $\tan \theta$ (C) $\sin \theta$ (D) $\cos \theta$
19. If $\tan x + \cot x = 2$, then $\sin^{2n} x + \cos^{2n} x$ is equal to :
 (A) 2^n (B) $-\frac{1}{2}$ (C) $\frac{1}{2}$ (D) None of these
20. The function f given by $f(x) = \sin\left(\frac{\pi x}{2}\right) + 2 \cos\left(\frac{\pi x}{3}\right) - \tan\left(\frac{\pi x}{3}\right)$ is periodic with period :
 (A) 6 (B) 3 (C) 4 (D) 12
21. If $P_n = \cos^n \theta + \sin^n \theta$ then $P_n - P_{n-2} = kP_{n-4}$, where $k =$
 (A) 1 (B) $-\sin^2 \theta \cos^2 \theta$ (C) $\sin^2 \theta$ (D) $\cos^2 \theta$
22. If $\sin \theta + \cos \theta = 2$ then the value of $\sin^8 \theta + \cos^8 \theta$ is equal to :
 (A) 2 (B) 2^8 (C) 2^4 (D) None of these
23. $\cos(2001)\pi + \cot(2001)\frac{\pi}{2} + \sec(2001)\frac{\pi}{3} + \tan(2001)\frac{\pi}{4} + \operatorname{cosec}(2001)\frac{\pi}{6}$ equals to :
 (A) 0 (B) 1 (C) -2 (D) Not defined
24. If $2 \tan \frac{\alpha}{2} = \tan \frac{\beta}{2}$, then $\frac{3+5 \cos \beta}{5+3 \cos \beta}$ is equal to :
 (A) $\cos \alpha$ (B) $\cos \beta$ (C) $\sin \alpha$ (D) $\sin \beta$
25. The expression $4 \cos^4 x - 2 \cos 2x - \frac{1}{2} \cos 4x$ when simplified reduces to :
 (A) $\frac{2}{3}$ (B) $\frac{3}{2}$ (C) $-\frac{2}{3}$ (D) $-\frac{3}{2}$

*26. If $\sin \theta + \sin \phi = a$ and $\cos \theta + \cos \phi = b$, then :

- (A) $\cos\left(\frac{\theta - \phi}{2}\right) = \pm \frac{1}{2} \sqrt{a^2 + b^2}$ (B) $\cos\left(\frac{\theta - \phi}{2}\right) = \pm \sqrt{a^2 - b^2}$
 (C) $\cos\left(\frac{\theta - \phi}{2}\right) = \pm \sqrt{\frac{4 - a^2 - b^2}{a^2 + b^2}}$ (D) $\cos(\theta - \phi) = \frac{a^2 + b^2 - 2}{2}$

*27. If $\sin(x - y) = \cos(x + y) = \frac{1}{2}$ then the values of x and y lying between 0 and π are given by :

- (A) $x = \frac{\pi}{4}, y = \frac{3\pi}{4}$ (B) $x = \frac{\pi}{4}, y = \frac{\pi}{12}$ (C) $x = \frac{5\pi}{4}, y = \frac{5\pi}{12}$ (D) $x = \frac{11\pi}{12}, y = \frac{3\pi}{4}$

28. If $\tan \theta \cdot \tan \phi = \sqrt{\frac{(a-b)}{(a+b)}}$, then $(a - b \cos 2\theta)(a - b \cos 2\phi) =$

- (A) a^2 (B) b^2 (C) $a^2 + b^2$ (D) $a^2 - b^2$

29. If $\sin x + \cos x + \tan x + \cot x + \sec x + \csc x = 7$ and $\sin 2x = a - b\sqrt{7}$, then ordered pair (a, b) can be :

- (A) (6, 2) (B) (8, 3) (C) (22, 8) (D) (11, 4)

30. A quadratic equation whose roots are $\sin^2 18^\circ, \cos^2 36^\circ$ is :

- (A) $16x^2 - 12x + 1 = 0$ (B) $x^2 - 12x + 1 = 0$
 (C) $16x^2 - 12x - 1 = 0$ (D) None of these

31. If $D = \begin{vmatrix} 1 & \cos \theta & 1 \\ -\sin \theta & 1 & -\cos \theta \\ -1 & \sin \theta & 1 \end{vmatrix}$ then D lies in the interval :

- (A) $[0, 4]$ (B) $[2, 4]$ (C) $[2 - \sqrt{2}, 2 + \sqrt{2}]$ (D) $[-2, 2]$

32. $\sin x + 2 \sin 2x = 3 + \sin 3x, 0 \leq x \leq 2\pi$ has :

- (A) 2 solutions in I quadrant (B) one solution in II quadrant
 (C) no solution in any quadrant (D) one solution in each quadrant

33. Minimum value of $(\sin \theta + \csc \theta)^2 + (\cos \theta + \sec \theta)^2$ is :

- (A) 0 (B) 2 (C) 7 (D) 9

34. $\frac{1}{2} \tan \frac{\alpha}{2} + \frac{1}{4} \tan \frac{\alpha}{4} + \frac{1}{8} \tan \frac{\alpha}{8} + \dots + \frac{1}{2^n} \tan \frac{\alpha}{2^n} + \cot \alpha =$

- (A) $\frac{1}{2^n} \tan \frac{\alpha}{2^n}$ (B) $\frac{1}{2^n} \cot \frac{\alpha}{2^n}$ (C) $\frac{1}{2^{n-1}} \tan \frac{\alpha}{2^n}$ (D) $\frac{1}{2^{n-1}} \cot \frac{\alpha}{2^n}$

35. If α is a root of $25 \cos^2 \theta + 5 \cos \theta - 12 = 0$, $\frac{\pi}{2} < \alpha < \pi$, then $\sin 2\alpha$ is equal to :

- (A) $\frac{24}{25}$ (B) $-\frac{24}{25}$ (C) $\frac{13}{18}$ (D) $-\frac{13}{18}$

36. If $x = y \cos \frac{2\pi}{3} = z \cos \frac{4\pi}{3}$, then $xy + yz + zx =$

- (A) -1 (B) 0 (C) 1 (D) 2

37. If the expression

$$\cos\left(x - \frac{3\pi}{2}\right) + \sin\left(\frac{3\pi}{2} + x\right) + \sin(32\pi + x) - 18\cos(19\pi - x) + \cos(56\pi + x) - 9\sin(x + 17\pi)$$

is expressed in the form of $a \sin x + b \cos x$, then $(a + b)$ is equal to :

- (A) 17 (B) 27 (C) 13 (D) 23

38. Let A and B denote the statements A : $\cos \alpha + \cos \beta + \cos \gamma = 0$ and B : $\sin \alpha + \sin \beta + \sin \gamma = 0$.

If $\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta) = -\frac{3}{2}$, then :

- (A) A is false and B is true (B) Both A and B are true
(C) Both A and B are false (D) A is true and B is false

39. The number of $x \in [0, 2\pi]$ for which $\left| \sqrt{2\sin^4 x + 18\cos^2 x} - \sqrt{2\cos^4 x + 18\sin^2 x} \right| = 1$ is :

- (A) 2 (B) 4 (C) 6 (D) 8

40. Let $f(x) = \sin^4 x + \cos^4 x$. Then f is an increasing function in the interval :

- (A) $\left[0, \frac{\pi}{4}\right]$ (B) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (C) $\left[\frac{\pi}{2}, \frac{5\pi}{8}\right]$ (D) $\left[\frac{5\pi}{8}, \frac{3\pi}{4}\right]$

41. If $A > 0$, $B > 0$ and $A + B = \frac{\pi}{6}$, then the minimum value of $\tan A + \tan B$ is :

- (A) $\sqrt{3} - \sqrt{2}$ (B) $2 - \sqrt{3}$ (C) $4 - 2\sqrt{3}$ (D) $\frac{2}{\sqrt{3}}$

42. The angle of elevation of the top of a vertical tower from a point A, due east of it is 45° . The angle of elevation of the top of the same tower a point B, due south of A is 30° . If the distance between A and B is $54\sqrt{2}$ m, then the height of the tower (in meters), is :

- (A) $36\sqrt{3}$ (B) 54 (C) $54\sqrt{3}$ (D) 180

43. In a $\triangle ABC$, $\frac{a}{b} = 2 + \sqrt{3}$ and $\angle C = 60^\circ$. Then the ordered pair $(\angle A, \angle B)$ is equal to :

- (A) $(15^\circ, 105^\circ)$ (B) $(105^\circ, 15^\circ)$ (C) $(45^\circ, 75^\circ)$ (D) $(75^\circ, 45^\circ)$

44. If $f(x) = 2 \tan^{-1} x + \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $x > 1$, then $f(5)$ is equal to :

- (A) $\frac{\pi}{2}$ (B) π (C) $4 \tan^{-1}(5)$ (D) $\tan^{-1}\left(\frac{65}{156}\right)$

45. If $2 \cos \theta + \sin \theta = 1$ $\left(\theta \neq \frac{\pi}{2}\right)$, then $7 \cos \theta + 6 \sin \theta$ is equal to :

- (A) $1/2$ (B) 2 (C) $11/2$ (D) $46/5$

46. Let 10 vertical poles standing at equal distances on a straight line, subtend the same angle of elevation α at a point O on this line and all the poles are on the same side of O. If the height of the longest pole is 'h' and the distance of the foot of the smallest pole from O is 'a'; then the distance between two consecutive poles, is :

- (A) $\frac{h \sin \alpha + a \cos \alpha}{9 \sin \alpha}$ (B) $\frac{h \cos \alpha - a \sin \alpha}{9 \cos \alpha}$ (C) $\frac{h \cos \alpha + a \sin \alpha}{9 \sin \alpha}$ (D) $\frac{h \sin \alpha + a \cos \alpha}{9 \cos \alpha}$

47. Let f be an odd function defined on the set of real numbers such that for $x \geq 0$, $f(x) = 3\sin x + 4\cos x$.
Then $f(x)$ at $x = -\frac{11\pi}{6}$ is equal to :
- (A) $\frac{3}{2} + 2\sqrt{3}$ (B) $-\frac{3}{2} + 2\sqrt{3}$ (C) $\frac{3}{2} - 2\sqrt{3}$ (D) $-\frac{3}{2} - 2\sqrt{3}$
48. The angle of elevation of the top of a vertical tower at a point P on the horizontal ground was observed to be α . After moving a distance 2 meters from P towards the foot of the tower, the angle of elevation changes to β . Then the height (in meters) of the tower is :
- (A) $\frac{2\sin\alpha\sin\beta}{\sin(\beta-\alpha)}$ (B) $\frac{\sin\alpha\sin\beta}{\cos(\beta-\alpha)}$ (C) $\frac{2\sin(\beta-\alpha)}{\sin\alpha\sin\beta}$ (D) $\frac{\cos(\beta-\alpha)}{\sin\alpha\sin\beta}$
49. The number of solutions of the equation $\sin 2x - 2\cos x + 4\sin x = 4$ in the interval $[0, 5\pi]$ is :
- (A) 3 (B) 5 (C) 4 (D) 6
50. **Statement I :** The equation $(\sin^{-1}x)^3 + (\cos^{-1}x)^3 - a\pi^3 = 0$ has a solution for all $a \geq \frac{1}{32}$.
Statement II : For any $x \in R$, $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ and $0 \leq \left(\sin^{-1}x - \frac{\pi}{4}\right)^2 \leq \frac{9\pi^2}{16}$.
- (A) Both statements I and II are false (B) Statement I is false and statement II is true
(C) Both statements I and II are true (D) Statement I is true and statement II is false
51. The function $f(x) = |\sin 4x| + |\cos 2x|$, is a periodic function with period :
- (A) 2π (B) π (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$
52. The principal value of $\tan^{-1}\left(\cot \frac{43\pi}{4}\right)$ is :
- (A) $-\frac{3\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $-\frac{\pi}{4}$ (D) $\frac{\pi}{4}$
53. A value of x for which $\sin\left(\cot^{-1}(1+x)\right) = \cos\left(\tan^{-1}x\right)$ is :
- (A) $-\frac{1}{2}$ (B) 1 (C) 0 (D) $\frac{1}{2}$
54. The number of solutions of the equation, $\sin^{-1}x = 2\tan^{-1}x$ (in principal values) is :
- (A) 1 (B) 4 (C) 2 (D) 3
55. **Statement 1 :** The number of common solutions of the trigonometric equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta - 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is two :
Statement 2 : The number of solutions of the equation, $2\cos^2\theta - 3\sin\theta = 0$ in the interval $[0, \pi]$ is two.
- (A) Statement 1 is true; Statement 2 is true; Statement 2 is a correct explanation for Statement 1
(B) Statement 1 is true; Statement 2 is true; Statement 2 is not a correct explanation for Statement 1
(C) Statement 1 is true; Statement 2 is false
(D) Statement 1 is false; Statement 2 is true

56. If $S = \tan^{-1}\left(\frac{1}{n^2 + n + 1}\right) + \tan^{-1}\left(\frac{1}{n^2 + 3n + 3}\right) + \dots + \tan^{-1}\left(\frac{1}{1 + (n+19)(n+20)}\right)$, then $\tan S$ is equal to :
- (A) $\frac{20}{401 + 20n}$ (B) $\frac{n}{n^2 + 20n + 1}$ (C) $\frac{20}{n^2 + 20n + 1}$ (D) $\frac{n}{401 + 20n}$
57. If m and M are the respective minimum and the maximum values of $4 + \frac{1}{2}\sin^2 2x - 2\cos^4 x$, $x \in R$, then $M - m$ is equal to :
- (A) $15/4$ (B) $9/4$ (C) $7/4$ (D) $1/4$
58. Let $P = \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$ and $Q = \{\theta : \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$ be two sets. Then :
- (A) $P \subset Q$ and $Q - P \neq \phi$ (B) $Q \not\subset P$
 (C) $P \not\subset Q$ (D) $P = Q$
59. If $\cos \alpha + \cos \beta = \frac{3}{2}$ and $\sin \alpha + \sin \beta = \frac{1}{2}$ and θ is the arithmetic mean of α and β , then $\sin 2\theta + \cos 2\theta$ is equal to :
- (A) $3/5$ (B) $4/5$ (C) $7/5$ (D) $8/5$
60. The number of values of α in $[0, 2\pi]$ for which $2\sin^3 \alpha - 7\sin^2 \alpha + 7\sin \alpha = 2$, is :
- (A) 6 (B) 4 (C) 3 (D) 1
61. If $\operatorname{cosec} \theta = \frac{p+q}{p-q}$ ($p \neq q \neq 0$), then $\left| \cot\left(\frac{\pi}{4} + \frac{\theta}{2}\right) \right|$ is equal to :
- (A) $\sqrt{\frac{p}{q}}$ (B) $\sqrt{\frac{q}{p}}$ (C) $\sqrt{pq}$ (D) pq

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

62. If $(1 + \tan 1^\circ)(1 + \tan 2^\circ)(1 + \tan 3^\circ) \dots (1 + \tan 45^\circ) = 2^n$, then ' n ' is equal to _____.
63. If $P = \cos \frac{\pi}{20} \cdot \cos \frac{3\pi}{20} \cdot \cos \frac{7\pi}{20} \cdot \cos \frac{9\pi}{20}$ and $Q = \cos \frac{\pi}{11} \cdot \cos \frac{2\pi}{11} \cdot \cos \frac{4\pi}{11} \cdot \cos \frac{8\pi}{11} \cdot \cos \frac{16\pi}{11}$, then $\frac{P}{Q}$ is:
64. If A, B, C, D are the angles of a cyclic quadrilateral, then $\cos A + \cos B + \cos C + \cos D =$
65. The function f given by $f(x) = \sin\left(\frac{\pi x}{2}\right) + 2\cos\left(\frac{\pi x}{3}\right) - \tan\left(\frac{\pi x}{3}\right)$ is periodic with period:
66. If $\sin \theta + \operatorname{cosec} \theta = 2$ then the value of $\sin^8 \theta + \operatorname{cosec}^8 \theta$ is equal to _____.
67. Minimum value of $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2$ is:
68. If $x = y \cos \frac{2\pi}{3} = z \cos \frac{4\pi}{3}$, then $xy + yz + zx =$

69. If the expression

$$\cos\left(x - \frac{3\pi}{2}\right) + \sin\left(\frac{3\pi}{2} + x\right) + \sin(32\pi + x) - 18\cos(19\pi - x) + \cos(56\pi + x) - 9\sin(x + 17\pi)$$
 is expressed in the form of $a \sin x + b \cos x$, then $(a + b)$ is equal to _____.
70. Range of $\log_{\sqrt{5}} \left[\sqrt{2}(\sin x - \cos x) + 3 \right]$ is $[a, b]$
 then $a + b$ is:
71. The angle of elevation of the top of a vertical tower from a point A , due east of it is 45° . The angle of elevation of the top of the same tower a point B , due south of A is 30° . If the distance between A and B is $54\sqrt{2} \text{ m}$, then the height of the tower (in meter), is:
72. If $2\cos\theta + \sin\theta = 1\left(\theta \neq \frac{\pi}{2}\right)$, then $7\cos\theta + 6\sin\theta$ is equal to _____.
73. The number of solutions of the equation $\sin 2x - 2\cos x + 4\sin x = 4$ in the interval $[0, 5\pi]$ is:
74. The number of solutions of the equation, $\sin^{-1} x = 2\tan^{-1} x$ (in principal values) is:
75. If $\cos\alpha + \cos\beta = \frac{3}{2}$ and $\sin\alpha + \sin\beta = \frac{1}{2}$ and θ is the arithmetic mean of α and β , then
 $(\sin 2\theta + \cos 2\theta) \times 5$ is equal to _____.
76. The number of value of α in $[0, 2\pi]$ for which $2\sin^3\alpha - 7\sin^2\alpha + 7\sin\alpha = 2$, is:

Conic Section

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT.

- Which one of the following equations represented in parametrically, represents equation to a parabolic profile ?
 (A) $x = 3\cos t, y = 4\sin t$ (B) $x^2 - 2 = -2\cos t, y = 4\cos^2 \frac{t}{2}$
 (C) $\sqrt{x} = \tan t, \sqrt{y} = \sec t$ (D) $x = \sqrt{1 - \sin t}, y = \sin \frac{t}{2} + \cos \frac{t}{2}$
- The length of latusrectum of the parabola $25[(x-2)^2 + (y-3)^2] = (3x-4y+7)^2$ is :
 (A) 4 (B) 2 (C) $\frac{1}{5}$ (D) $\frac{2}{5}$

PARAGRAPH FOR QUESTIONS 3 - 6

The equation of conic given by : $S : 9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$.

- The curve S represents a(an) :
 (A) circle (B) parabola (C) ellipse (D) hyperbola
- The axis of conic is :
 (A) $3x - 4y = 0$ (B) $3x + 4y = 0$ (C) $3x - 4y + 12 = 0$ (D) $4x + 3y + 12 = 0$
- The vertex of conic is :
 (A) $\left(\frac{48}{25}, \frac{36}{25}\right)$ (B) $\left(-\frac{43}{25}, -\frac{129}{100}\right)$ (C) $\left(-\frac{48}{25}, -\frac{36}{25}\right)$ (D) $\left(\frac{43}{25}, \frac{129}{100}\right)$
- The directrix of curve is :
 (A) $4x + 3y + \frac{53}{4} = 0$ (B) $4x - 3y + 12 = 0$ (C) $4x + 3y + 12 = 0$ (D) $3x - 4y = 0$
- If $(2, -8)$ is one end of a focal chord of the parabola $y^2 = 32x$, then the other end of the focal chord is :
 (A) $(32, 32)$ (B) $(32, -32)$ (C) $(-2, 8)$ (D) $(2, 8)$
- The locus of feet of perpendiculars drawn from vertex of the parabola $y^2 = 4ax$ upon all such chords of the parabola which subtends a right angle at the vertex, is :
 (A) $x^2 + y^2 - 4ax = 0$ (B) $x^2 + y^2 - 2ax = 0$
 (C) $x^2 + y^2 + 2ax = 0$ (D) $x^2 + y^2 + 4ax = 0$
- The locus of trisection point of any arbitrary double ordinate of the parabola $x^2 = 4by$ is :
 (A) $9x^2 = by$ (B) $3x^2 = 2by$ (C) $9x^2 = 4by$ (D) $9x^2 = 2by$
- Let A and B be two points on a parabola $y^2 = x$ with vertex V such that VA is perpendicular to VB and θ is the angle between the chord VA and the axis of the parabola. The value of $\frac{|VA|}{|VB|}$ is :
 (A) $\tan \theta$ (B) $\tan^3 \theta$ (C) $\cot^2 \theta$ (D) $\cot^3 \theta$
- The locus of vertices of the family parabola $y = \frac{a^3 x^2}{3} + \frac{a^2 x}{2} - 2a$ is :
 (A) $xy = \frac{3}{4}$ (B) $xy = \frac{35}{16}$ (C) $xy = \frac{64}{105}$ (D) $xy = \frac{105}{64}$

12. The line $4x - 7y + 10 = 0$ intersects the parabola, $y^2 = 4x$ at the point A and B. The coordinates of point x of intersection of the tangents drawn at the point A and B are :
- (A) $\left(\frac{7}{5}, \frac{5}{2}\right)$ (B) $\left(-\frac{5}{7}, -\frac{7}{2}\right)$ (C) $\left(\frac{5}{7}, \frac{7}{2}\right)$ (D) $\left(-\frac{7}{2}, -\frac{5}{2}\right)$
13. The locus of foot of the perpendiculars drawn from the vertex on a variable tangent to the parabola $y^2 = 4ax$ is :
- (A) $x(x^2 + y^2) + ay^2 = 0$ (B) $y(x^2 + y^2) + ax^2 = 0$
 (C) $x(x^2 - y^2) + ay^2 = 0$ (D) None of these
14. AP and BP are tangents to the parabola, $y^2 = 4x$ at A and B. If the chord AB passes through a fixed point $(+1, +1)$, then the equation of locus of P is :
- (A) $y = 2(x - 1)$ (B) $y^2 = 2(x - 1)$ (C) $y = 2(x + 1)$ (D) $y^2 = 2(x + 1)$
15. AB, AC are tangents to a parabola $y^2 = 4ax$, p_1 , p_2 and p_3 are the lengths of the perpendiculars from A, B and C respectively on any tangent to the curve, then p_2 , p_1 , p_3 are in :
- (A) AP (B) GP (C) HP (D) None of these
16. At any point P on the parabola $y^2 - 2y - 4x + 5 = 0$ a tangent is drawn which meets the directrix at Q. Then, the locus of R which divides QP externally in the ratio $\frac{1}{2} : 1$ is :
- (A) $(x + 1)(y - 1)^2 = 4$ (B) $(x + 1)(y - 1)^2 + 4 = 0$
 (C) $(x - 1)(y + 1)^2 + 4 = 0$ (D) $(x - 1)(y + 1)^2 + 4 = 0$
17. The normal at the point $P(ap^2, 2ap)$ meets the parabola $y^2 = 4ax$ again at $Q(aq^2, 2aq)$ such that the line joining the origin to P and Q are at the right angle. Then, p^2 is equal to :
- (A) 1 (B) 2 (C) $\sqrt{2}$ (D) 4
18. The ordinates of points P and Q on the parabola $y^2 = 12x$ are in the ratio 1 : 2. Then, the locus of the point of intersection of the normals to the parabola at P and Q is :
- (A) $x = 3 \left[7 \left(\frac{y}{18} \right)^{2/3} + 2 \right]$ (B) $x = \left(\frac{y}{18} \right)^{2/3} + 2$
 (C) $x = 3 \left[7 \left(\frac{y}{18} \right)^{2/3} + 1 \right]$ (D) None of these
19. A normal is drawn to parabola $y^2 = 4ax$ at any point other than the vertex. If it cuts the parabola again at a point whose distance from the vertex is not less than :
- (A) $4a$ (B) $4\sqrt{6}a$ (C) $2\sqrt{6}a$ (D) $3\sqrt{6}a$

20. The locus of the point of intersection of normals at the ends of parallel chords of gradient m of the parabola $y^2 = 4ax$, is :
- (A) $2xm^2 - ym^3 = 4a(2 + m^2)$ (B) $2xm^2 + ym^3 = 4a(2 + m^2)$
 (C) $2xm + ym^2 = 4a(2 + m)$ (D) $2xm^2 + ym^3 = 4a(2 - m^2)$
21. The length of intercept on the normal at the point $(at^2, 2at)$ of the parabola $y^2 = 4ax$ made by circle which is described on a focal distance of the given point as diameter is :
- (A) $a\sqrt{1+t}$ (B) $a\sqrt{1+2t}$ (C) $a\sqrt{1+t^2}$ (D) $a\sqrt{1+t^4}$
22. The two parabolas $y^2 = 4ax$ and $y^2 = 4c(x-b)$ cannot have a common normal other than the axes, unless :
- (A) $\frac{b}{a-c} > 2$ (B) $\frac{a}{b-c} > 2$ (C) $\frac{b}{a-c} < 2$ (D) $\frac{a}{b-c} < 2$
23. Tangents are drawn at the point, where the line $lx + my + n = 0$ is intersected by the parabola $y^2 = 4ax$, then the point of intersection of tangents is :
- (A) $\left(\frac{n}{\ell}, -\frac{2am}{\ell}\right)$ (B) $\left(-\frac{n}{\ell}, -\frac{2am}{\ell}\right)$ (C) $\left(-\frac{n}{\ell}, -\frac{2am}{\ell}\right)$ (D) None of these
24. The locus of the poles of focal chords of the parabola $y^2 = 4ax$ is :
- (A) the tangent at vertex (B) $x = -2a$
 (C) the directrix (D) $x = y + a$
25. Minimum distance between the parabolas $y^2 - 4x - 8y + 40 = 0$ and $x^2 - 8x - 4y + 40 = 0$ is :
- (A) 0 (B) $\sqrt{3}$ (C) $2\sqrt{2}$ (D) $\sqrt{2}$
26. The set of values of m for which a chord of slope m of the circle $x^2 + y^2 = 4$ touches the parabola $y^2 = 4x$, is :
- (A) $\left(-\infty, -\sqrt{\frac{\sqrt{2}-1}{2}}\right) \cup \left(\sqrt{\frac{\sqrt{2}-1}{2}}, \infty\right)$ (B) $(-\infty, 1) \cup (1, \infty)$
 (C) $(-1, 1)$ (D) $\mathbb{R}$
27. The set of values of h for which the number of distinct common normals of $(x-2)^2 = 4(y-3)$ and $x^2 + y^2 - 2x - hy - c = 0$, ($c > 0$) is 3, then :
- (A) $(2, \infty)$ (B) $(4, \infty)$ (C) $(2, 4)$ (D) $(10, \infty)$
28. The equation $2x^2 + 3y^2 - 8x - 18y + 35 = k$ represents :
- (A) no locus, if $k > 0$ (B) an ellipse, if $k < 0$
 (C) a point, if $k = 0$ (D) a hyperbola, if $k > 0$
29. The eccentric angle of the extremities of latusrecta of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is :
- (A) $\tan^{-1}\left(\pm \frac{b}{ae}\right)$ (B) $\tan^{-1}\left(\pm \frac{a}{be}\right)$ (C) $2\tan^{-1}\left(\pm \frac{b}{ae}\right)$ (D) None of these

30. If any two chords can be drawn through two points on the major axis of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ equidistant from the centre, then the value of $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} \cdot \tan \frac{\gamma}{2} \cdot \tan \frac{\delta}{2}$ is : (where $\alpha, \beta, \gamma, \delta$ are the eccentric angles of extremities of the chords)
- (A) 0 (B) 1 (C) -1 (D) $\frac{1}{2}$
31. The equations of tangents to the ellipse $4x^2 + 3y^2 = 5$, which are inclined at an angle of 60° to the X-axis is :
- (A) $y = \sqrt{3} \pm \sqrt{\frac{65}{12}}$ (B) $y = \sqrt{3}x \pm 5$ (C) $y = \frac{1}{\sqrt{3}}x \pm \sqrt{\frac{65}{12}}$ (D) $y = \frac{1}{\sqrt{3}}x \pm 5$
32. The tangents drawn to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = a + b$ at the points, where it cut by the tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ subtends an angle of :
- (A) 60° (B) 30° (C) 45° (D) 90°
33. The length of normal (terminated by the major axis) at a point of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is :
- (A) $\frac{b}{a}(r + r_1)$ (B) $\frac{b}{a}|r - r_1|$ (C) $\frac{b}{a}\sqrt{rr_1}$ (D) independent of r, r_1
34. If ω is one of the angles between the normals to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the points whose eccentric angles are θ and $\frac{\pi}{2} + \theta$, then $\frac{2\cot \omega}{\sin 2\theta}$ is :
- (A) $\frac{e^2}{1-e^2}$ (B) $\frac{e}{1-e}$ (C) $\frac{e^2}{\sqrt{1-e^2}}$ (D) $\frac{e^2}{1-e}$
35. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the X-axis at Q. If M is the mid-point of the segment PQ, then the locus of M intersects the latusrectum of the given ellipse at the points:
- (A) $\left(\pm \frac{3\sqrt{5}}{2}, \pm \frac{2}{7}\right)$ (B) $\left(\pm \frac{3\sqrt{5}}{2}, \pm \frac{\sqrt{19}}{4}\right)$ (C) $\left(\pm 2\sqrt{3}, \pm \frac{1}{7}\right)$ (D) $\left(\pm 2\sqrt{3}, \pm \frac{4\sqrt{3}}{7}\right)$
36. If the normal at an end of a latusrectum of an ellipse passes through one extremity of the minor axis, then the eccentricity of ellipse is given by :
- (A) $e^4 - e^2 + 1 = 0$ (B) $e^4 + e^2 - 1 = 0$ (C) $e^2 + e - 1 = 0$ (D) None of these
37. A point on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at a distance equal to the mean of lengths of the semi-major axes semi-minor axes from the centre is:
- (A) $\left(\frac{2\sqrt{91}}{7}, \frac{3\sqrt{105}}{14}\right)$ (B) $\left(\frac{2\sqrt{91}}{7}, -\frac{3\sqrt{91}}{14}\right)$
- (C) $\left(-\frac{2\sqrt{105}}{7}, -\frac{3\sqrt{91}}{14}\right)$ (D) None of these
38. A tangent to the ellipse $4x^2 + 9y^2 = 36$ is cut by the tangent at the extremities of the major axis at T and T' . The circle on TT' as diameter passes through the point.
- (A) $(-\sqrt{5}, 0)$ (B) $(\sqrt{5}, 1)$ (C) $(0, 0)$ (D) $(3, 2)$

39. The eccentricity of an ellipse is $\frac{2}{3}$, latusrectum is 5 and centre (0, 0). The equation of ellipse is :
- (A) $\frac{x^2}{81} + \frac{y^2}{45} = 1$ (B) $\frac{4x^2}{81} + \frac{4y^2}{45} = 1$ (C) $\frac{x^2}{9} + \frac{y^2}{5} = 1$ (D) None of these
40. If $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$ and $x^2 - y^2 = c^2$ cut at right angles, then :
- (A) $a^2 + b^2 = 2c^2$ (B) $b^2 - a^2 = 2c^2$ (C) $a^2 - b^2 = 2c^2$ (D) $a^2 b^2 = 2c^2$
41. The value of m for which $y = mx + 6$ is a tangent to hyperbola $\frac{x^2}{100} - \frac{y^2}{49} = 1$, is :
- (A) $\sqrt{\frac{17}{20}}$ (B) $\sqrt{\frac{20}{17}}$ (C) $\sqrt{\frac{3}{20}}$ (D) $\sqrt{\frac{13}{20}}$
42. The area (in sq. units) of triangle formed by the lines $x - y = 0$, $x + y = 0$ and any tangent to the hyperbola $x^2 - y^2 = a^2$ is :
- (A) a^2 (B) $2a^2$ (C) $3a^2$ (D) $4a^2$
43. If $P(a \sec \alpha, b \tan \alpha)$ and $Q(a \sec \beta, b \tan \beta)$ are two points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Such that $\alpha - \beta = 2\theta$, then PQ touches the hyperbola:
- (A) $\frac{x^2}{a^2 \sec^2 \theta} - \frac{y^2}{b^2} = 1$ (B) $\frac{x^2}{a^2} - \frac{y^2}{b^2 \sec^2 \theta} = 1$ (C) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (D) None of these
44. The normal at P to a hyperbola of eccentricity e , intersects its transverse and conjugate axes at L and M respectively. If the locus of mid-point of LM is hyperbola, having eccentricity e_1 , then:
- (A) $e_1 = \frac{e}{e^2 - 1}$ (B) $e_1 = \sqrt{\frac{e}{e^2 - 1}}$ (C) $e_1 = \frac{e}{\sqrt{e^2 - 1}}$ (D) None of these
45. The line $lx + my + n = 0$ will be a normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $\frac{a^2}{l^2} - \frac{b^2}{m^2}$ is equal to :
- (A) $\frac{a^2 + b^2}{n^2}$ (B) $\frac{a^2 - b^2}{n^2}$ (C) $\frac{(a^2 + b^2)^2}{n^2}$ (D) $\frac{(a^2 - b^2)^2}{n^2}$
46. The locus of feet of the normals drawn from a point to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ lie on the circle:
- (A) $a^2 y(x + h) + b^2 x(y + k) = 0$ (B) $b^2 y(x + h) + a^2 x(y + k) = 0$
 (C) $a^2 y(x - h) + b^2 x(y - k) = 0$ (D) $b^2 y(x - h) + a^2 x(y - k) = 0$
47. If normal to the rectangular hyperbola $xy = c^2$ at the point t on it intersect the hyperbola at t_1 , then $t^3 t_1$, is:
- (A) 1 (B) 2 (C) -1 (D) -2
48. The locus of point of intersection of tangents at the ends normal chord of the hyperbola $x^2 - y^2 = a^2$ is:
- (A) $y^4 - x^4 = 4a^2 x^2 y^2$ (B) $y^2 - x^2 = 4a^2 x^2 y^2$
 (C) $a^2(y^2 - x^2) = 4x^2 y^2$ (D) $y^2 + x^2 = 4a^2 x^2 y^2$

49. If the normal at the point $p(x_i, y_i), i = 1, 2, 3, 4$ on the hyperbola $xy = c^2$ are concurrent at the point $\theta(h, k)$ then $\frac{(x_1 + x_2 + x_3 + x_4)(y_1 + y_2 + y_3 + y_4)}{x_1 x_2 x_3 x_4}$ is :
- (A) $\frac{hk}{c^2}$ (B) $\frac{hk}{c^4}$ (C) $\frac{-hk}{c^2}$ (D) $\frac{-hk}{c^4}$
50. Let a and b respectively be the semi-transverse and semi-conjugate axes of a hyperbola whose eccentricity satisfies the equation $9e^2 - 18e + 5 = 0$. If $S(5, 0)$ is a focus and $5x = 9$ is the corresponding directrix of this hyperbola, then $a^2 - b^2$ is equal to :
- (A) 7 (B) -7 (C) 5 (D) -5
51. If the tangent at a point on the ellipse $\frac{x^2}{27} + \frac{y^2}{3} = 1$ meets the coordinate axes at A and B , and O is the origin, then the minimum area (in sq. units) of the triangle OAB is :
- (A) $\frac{9}{2}$ (B) $3\sqrt{3}$ (C) $9\sqrt{3}$ (D) 9
52. P and Q are two distinct points on the parabola, $y^2 = 4x$, with parameters t and t_1 respectively. If the normal at P passes through Q , then the minimum value of t_1^2 is :
- (A) 2 (B) 4 (C) 6 (D) 8
53. A hyperbola whose transverse axis is along the major axis of the conic, $\frac{x^2}{3} + \frac{y^2}{4} = 4$ and has vertices at the foci of this conic. If the eccentricity of the hyperbola is $3/2$ then which of the following points does NOT lie on it ?
- (A) $(0, 2)$ (B) $(\sqrt{5}, 2\sqrt{2})$ (C) $(\sqrt{10}, 2\sqrt{3})$ (D) $(5, 2\sqrt{3})$
54. An ellipse passes through the foci of the hyperbola, $9x^2 - 4y^2 = 36$ and its major and minor axes lie along the transverse and conjugate axes of the hyperbola respectively. If the product of eccentricities of the two conics is $\frac{1}{2}$, then which of the following points does not lie on the ellipse ?
- (A) $(\sqrt{13}, 0)$ (B) $\left(\frac{\sqrt{39}}{2}, \sqrt{3}\right)$ (C) $\left(\frac{1}{2}\sqrt{3}, \frac{\sqrt{3}}{2}\right)$ (D) $\left(\sqrt{\frac{13}{2}}, \sqrt{6}\right)$
55. If the distance between the foci of an ellipse is half the length of its latus rectum, then the eccentricity of the ellipse is :
- (A) $\frac{1}{2}$ (B) $\frac{2\sqrt{2}-1}{2}$ (C) $\sqrt{2}-1$ (D) $\frac{\sqrt{2}-1}{2}$
56. Let PQ be the double ordinate of the parabola, $y^2 = -4x$, where P lies in the second quadrant. If R divides PQ in the ratio $2 : 1$, then the locus of R is :
- (A) $9y^2 = 4x$ (B) $9y^2 = -4x$ (C) $3y^2 = 2x$ (D) $3y^2 = -2x$
57. If OB is the semi-minor axis of an ellipse, F_1 and F_2 are its foci and the angle between F_1B and F_2B is a right angle, then the square of the eccentricity of the ellipse is :
- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{2\sqrt{2}}$ (D) $\frac{1}{4}$

58. A stair-case of length l rests against a vertical wall and a floor of a room. Let P be a point on the stair-case, nearer to its end to on the wall, that divides its length in the ratio $1 : 2$. If the stair -case begins to slide on the floor, then the locus of P is :
- (A) an ellipse of eccentricity $\frac{1}{2}$ (B) an ellipse of eccentricity $\frac{\sqrt{3}}{2}$
 (C) a circle of radius $\frac{l}{2}$ (D) $\frac{\sqrt{3}}{2}l$
59. Let L_1 be the length of the common chords of the curves $x^2 + y^2 = 9$ and $y^2 = 8x$, and L_2 be the length of the latus rectum of $y^2 = 8x$, then :
- (A) $L_1 > L_2$ (B) $L_1 = L_2$ (C) $L_1 < L_2$ (D) $\frac{L_1}{L_2} = \sqrt{2}$
60. Let $P(3\sec\theta, 2\tan\theta)$ and $Q(3\sec\phi, 2\tan\phi)$ where $\theta + \phi = \frac{\pi}{2}$, be two distinct points on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$. Then the ordinate of the point of intersection of the normals at P and Q is :
- (A) $\frac{11}{3}$ (B) $-\frac{11}{3}$ (C) $\frac{13}{2}$ (D) $-\frac{13}{2}$
61. The minimum area of a triangle formed by any tangent to the ellipse $\frac{x^2}{16} + \frac{y^2}{81} = 1$ and the co-ordinate axes is:
- (A) 12 (B) 26 (C) 36 (D) 18
62. Two tangents are drawn from a point $(-2, -1)$ to the curve, $y^2 = 4x$. If α is the angle between them, then $|\tan\alpha|$ is equal to :
- (A) $\frac{1}{3}$ (B) 3 (C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$
63. A chord drawn through the focus of the parabola $y^2 = 6x$ such that its distance from the vertex of this parabola is $\frac{\sqrt{5}}{2}$, then its slope can be :
- (A) $\frac{\sqrt{5}}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{2}{\sqrt{5}}$ (D) $\frac{2}{\sqrt{3}}$
64. The tangent at an extremity (in the first quadrant) of latus rectum of the hyperbola $\frac{x^2}{4} - \frac{y^2}{5} = 1$, meets x -axis and y -axis at A and B respectively. Then $(OA)^2 - (OB)^2$, where O is the origin, equals :
- (A) $-\frac{20}{9}$ (B) $\frac{16}{9}$ (C) 4 (D) $-\frac{4}{3}$
65. Equation of the line passing through the points of intersection of the parabola $x^2 = 8y$ and the ellipse $\frac{x^2}{3} + y^2 = 1$ is :
- (A) $y - 3 = 0$ (B) $y + 3 = 0$ (C) $3y + 1 = 0$ (D) $3y - 1 = 0$

66. If a and c are positive real numbers and the ellipse $\frac{x^2}{4c^2} + \frac{y^2}{c^2} = 1$ has four distinct points in common with the circle $x^2 + y^2 = 9a^2$, then :
- (A) $9ac - 9a^2 - 2c^2 < 0$ (B) $6ac + 9a^2 - 2c^2 < 0$
 (C) $9ac - 9a^2 - 2c^2 > 0$ (D) $6ac + 9a^2 - 2c^2 > 0$
67. **Statement 1 :** The line $x - 2y = 2$ meets the parabola, $y^2 + 2x = 0$ only at the point $(-2, -2)$:
Statement 2 : The line $y = mx - \frac{1}{2m}$ ($m \neq 0$) is tangent to the parabola, $y^2 = -2x$ at the point $\left(-\frac{1}{2m^2}, -\frac{1}{m}\right)$.
- (A) Statement 1 is true; Statement 2 is false
 (B) Statement 1 is true; Statement 2 is true; Statement 2 is a correct explanation for Statement 1
 (C) Statement 1 is false; Statement 2 is true
 (D) Statement 1 is true; Statement 2 is true; Statement 2 is not a correct explanation for Statement 1
68. Let the equations of two ellipse be $E_1 : \frac{x^2}{3} + \frac{y^2}{2} = 1$ and $E_2 : \frac{x^2}{16} + \frac{y^2}{b^2} = 1$. If the product of their eccentricities is $\frac{1}{2}$, then the length of the minor axis of ellipse E_2 is :
- (A) 8 (B) 9 (C) 4 (D) 2
69. If the curves $\frac{x^2}{\alpha} + \frac{y^2}{4} = 1$ and $y^2 = 16x$ intersect at right angles, then a value of α is :
- (A) 2 (B) $4/3$ (C) $1/2$ (D) $3/4$
70. A tangent to the hyperbola $\frac{x^2}{4} - \frac{y^2}{2} = 1$ meets x -axis at P and y -axis at Q . Lines PR and QR are drawn such that $OPRQ$ is a rectangle (where O is the origin). Then R lies on :
- (A) $\frac{4}{x^2} + \frac{2}{y^2} = 1$ (B) $\frac{2}{x^2} - \frac{4}{y^2} = 1$ (C) $\frac{2}{x^2} + \frac{4}{y^2} = 1$ (D) $\frac{4}{x^2} - \frac{2}{y^2} = 1$
71. The point of intersection of the normals to the parabola $y^2 = 4x$ at the ends of its latus rectum is :
- (A) $(0, 2)$ (B) $(3, 0)$ (C) $(0, 3)$ (D) $(2, 0)$

Integer Answer Type Questions

The Answer to the following questions are positive integers of 1/2/3 digits and zero

72. If PQ is any focal chord of the parabola $y^2 = 32x$ and length of PQ can never be less than λ units, then λ is _____.
73. A tangent is drawn to the parabola $y^2 = 4x$ at the point ' P ' whose abscissa lies in the interval $[1, 4]$. If maximum possible area of the triangle formed by the tangent at ' P ', ordinate of the point ' P ' and the X -axis is λ sq. units, then λ is _____.

74. The normal at the ends of the latusrectum of the parabola $y^2 = 4x$ meet the parabola again at A and A' . If length $AA' = \lambda$ unit, then λ is _____.
75. If the orthocentre of the triangle formed by the points t_1, t_2, t_3 on the parabola $y^2 = 4ax$ is the focus, then the value of $|t_1t_2 + t_2t_3 + t_3t_1|$ is _____.
76. Two tangents are drawn from the point $(-2, -1)$ to the parabola $y^2 = 4x$. If θ is the angle between these tangents, then the value of $\tan \theta$ is _____.
77. If the distances of two points P and Q from the focus of a parabola $y^2 = 4x$ are 4 and 9 respectively, the distance of the point of intersection of tangents at P and Q from the focus is _____.
78. Let $f(x) = \begin{cases} a^x, & x < 2 \\ 8, & x = 2 \\ \frac{b(x^2 - b^2)}{x - 2}, & x > 2 \end{cases}$ If f is continuous at $x = 2$, then the locus of the pair of perpendicular tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = r^2$, then r^2 is _____.
79. If the ellipse $\frac{(x-h)^2}{M} + \frac{(y-k)^2}{N} = 1$ has major axis on the line $y = 2$, minor-axis on the line $x = -1$, major axis has length 10 and minor axis has length 4. Then, $h+k+M+N$ is _____.
80. If PQ is a focal chord of ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, which passes through $S(3, 0)$ and $PS = 2$, then length of PQ is _____.
81. Number of points on the ellipse $2x^2 + 5y^2 = 100$ from which pair of perpendicular tangents may be drawn to the ellipse $9x^2 + 16y^2 = 144$ is _____.
82. The length of the focal chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which makes an angle θ with the major axis is $\frac{\lambda ab^2}{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$, then the value of λ is _____.
83. The locus of the point of intersection of the lines $\sqrt{3}x - y - 4\sqrt{3}t = 0$ and $\sqrt{3}tx + ty - 4\sqrt{3} = 0$ (where t is a parameter) is a hyperbola whose eccentricity is _____.
84. The area of the triangle that a tangent at a point on the hyperbola $\frac{x^2}{16} - \frac{y^2}{a^2} = 1$ makes with its asymptotes is _____.
85. If abscissa of orthocentre of a triangle inscribed in a rectangular hyperbola $xy = 4$ is $\frac{1}{2}$, then the ordinate of orthocentre of triangle is _____.
86. The shortest distance between the curves $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $4x^2 + 4y^2 = a^2$ ($b > a$) is $f(a, b)$, then the value of $f(4, 6) + f(2, 3)$ is _____.